



MOON WALK

Walk with your own Rhythm



Qualcomm

AiAT

ARTIFICIAL INTELLIGENCE ASSOCIATION OF THAILAND



Moon Walk — Product Impact in Thailand

A clip-on cane that coaches optimal loading for faster sprain recovery

25%

of injuries are sprains —
#1 acute injury in Thai
sport

~70%

of ankle sprains recur
from poor loading
(global)

~\$30k

cost per ankle
sprain (global ED
benchmark)

Scarce therapists can't supervise daily loading. A low-cost smart cane delivers real-time optimal-loading guidance — cutting re-injury and speeding recovery.

Sources: SEA Games injury surveillance 2022 • Ankle sprain epidemiology • Thai PT workforce 2016 • Sports Health 2016

20.76 Days

Smeeming 2015 (Meta-analysis, 25 articles, Ankle Fracture). Early mobilization and weight-bearing returned patients to work and daily activities 20.76 days earlier (95% CI -40.02 to -1.50) with no increased complications (RR 1.22).

This is the hardest recovery-time number in the dataset.

North 2023 (n=42): Step count strongly predicts 1-year function (impact 0.817); absolute offloaded weight ranks low (0.309). Getting moving matters most.

Ribeiro-Castro 2023 (n=2255): Longer early walking bouts, tinner early walking bouts predict significantly better 90-day gait speed and symmetry.

Bottom Line Synthesis: Less offloading equates to ~3 weeks faster return to activity. **Honest Caveat:** Existing evidence is binary (immobilized vs. not). Stick Duty Factor as a graded, dosed variable has not yet been directly tested.

อาการดีขึ้นไหม

เมื่อไหร่ควรจะไปหาหมอ

เราควรลงน้ำหนักมากขึ้นหรือควรใช้ไม้เท้ามากขึ้น

Painpoint

Solution

DEMO

Architecture

Opportunity

Our Solution

MOON WALK

A clip-on AI module that attaches to any cane or walker — turning ordinary mobility aids into intelligent health companions.



The 3 Co-Equal Pillars of Moon Walk Recovery

Three distinct signals that characterize the limp, measure cane dependence, and track retraining—all calculated 100% relative to the patient's own baseline.



1. Symmetry & Rhythm (The Headline Limp Meter)

- Direct measure of L vs. R balance & steadiness.
- Tracking rhythm recovery & full symmetry.

How it's calculated:



Sensors: IMU only. Calculates plant interval ratio & consistency.



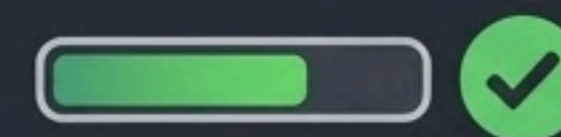
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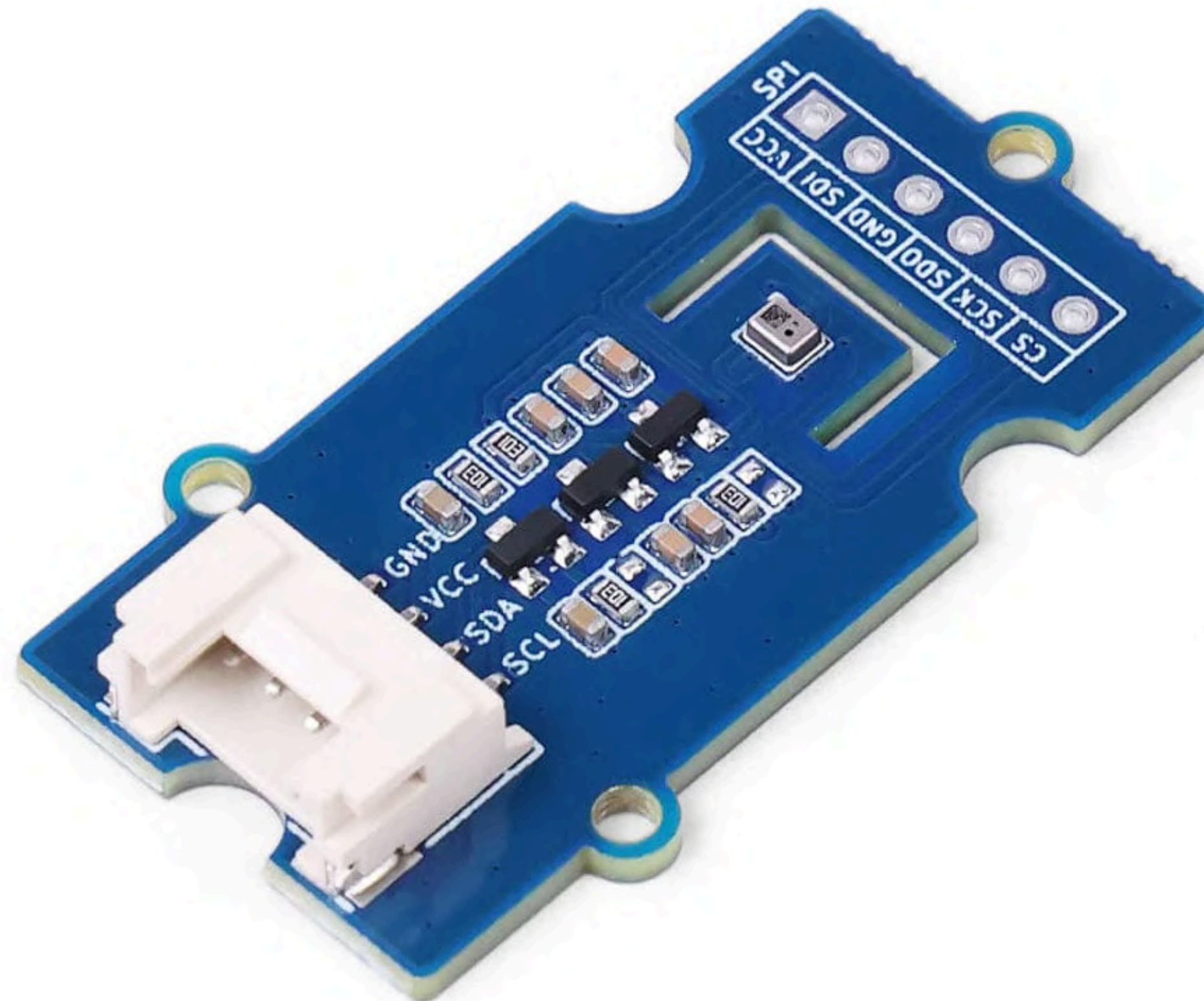
3. Session Training Load (The Daily "Did I Practice?" Score)

- 0-100 engagement score for total quality volume.
- Motivates progressive loading without overdoing it.

How it's calculated:



Sensors: Barometer + IMU. Scores & sums steps, mapped to a 0-100 log curve.



Bridge the gap with *Teamwork*

Build the product.
Reach the **moon**.



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Painpoint

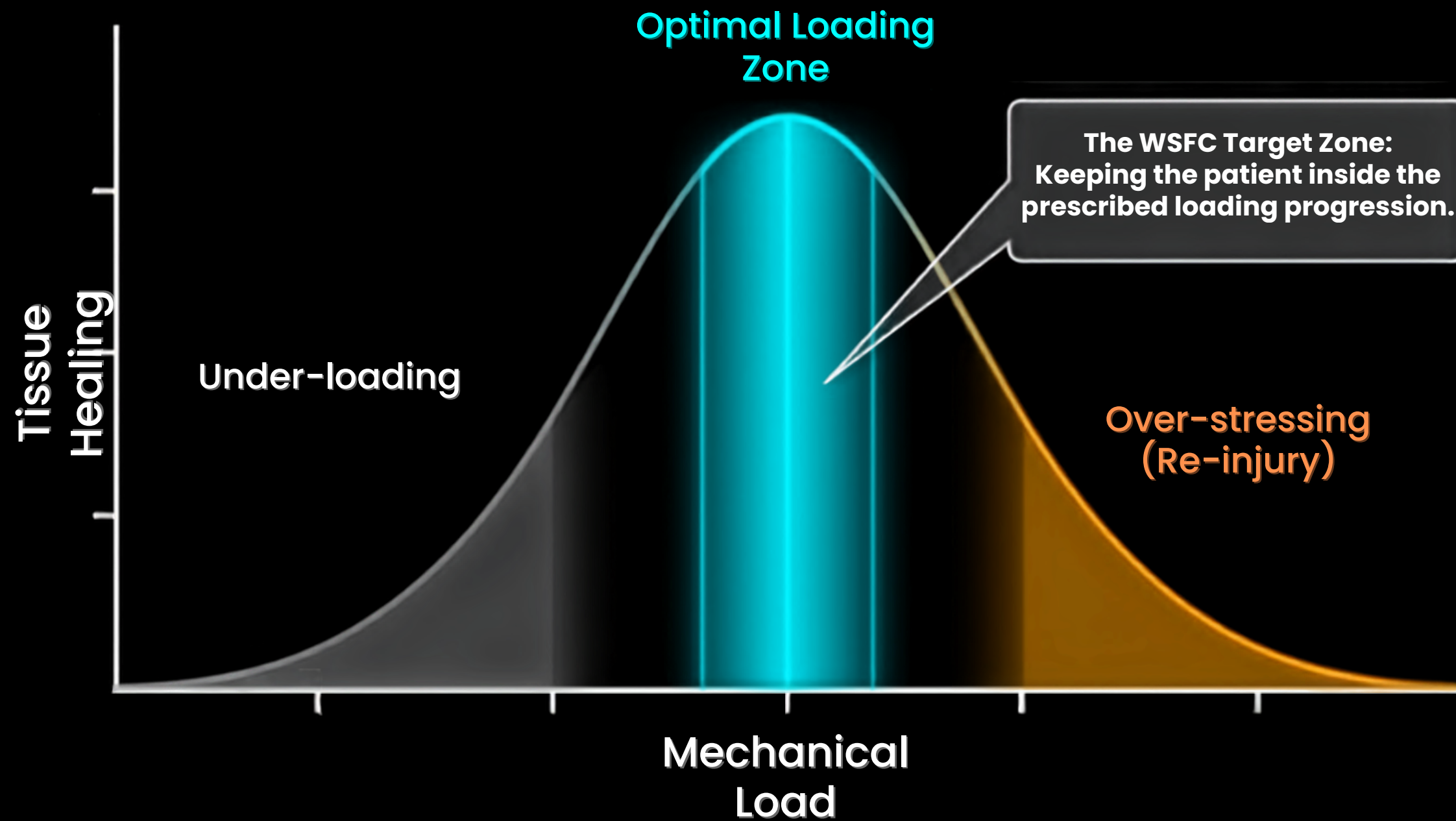
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The Healing Stimulus Curve



Bleakley CM, et al., 2012 (BJSM). Rest may be harmful; progressive load restores tissue strength.

Painpoint

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Metric 2 — Stick Duty Factor (Progressive Weaning)

A Single Stick Cycle (plant-to-plant)



What Improving It Means

The patient leans on the cane less, and for a shorter duration within each step. They are progressively reloading the limb, weaning off the support, and returning to a natural gait pattern.

The Mechanism

Planted duration divided by total cycle time. Validated by the barometer but executable via the IMU's measurement of angular-rate stillness. It measures the cane's loaded interval, not leg stance time.

DEMO



Painpoint

Solution

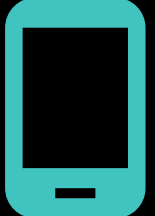
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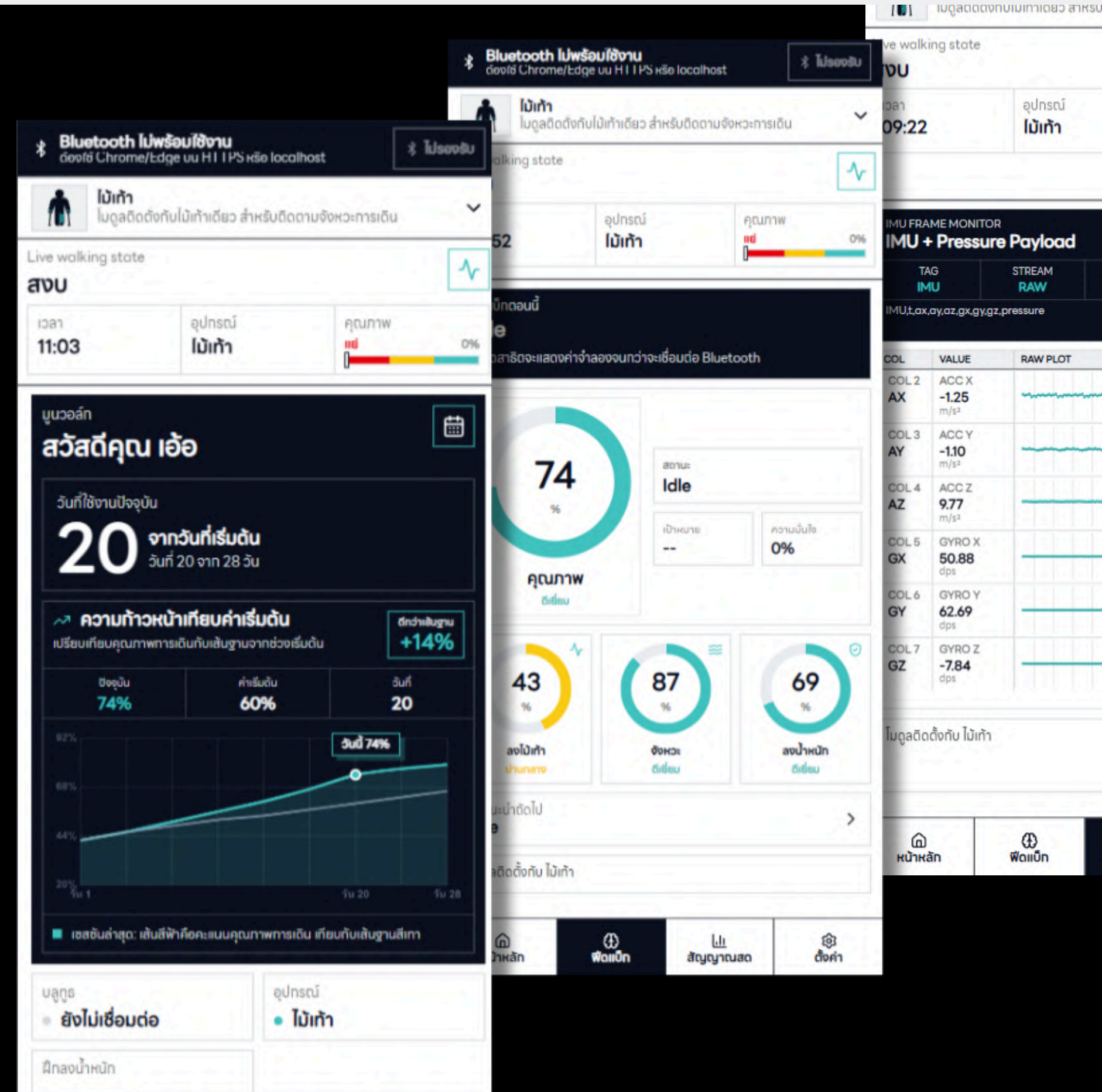
Architecture

Opportunity

Our Software

MOON
WALK

- 1 Walk 
- 2 Monitor 
- 3 Adjust
- 4 Get Better



Our Hardware

MOON WALK ARCHITECTURE

End-to-end system view based on our discussion: Sensor Node → Gateway → Processing → Storage → Analysis

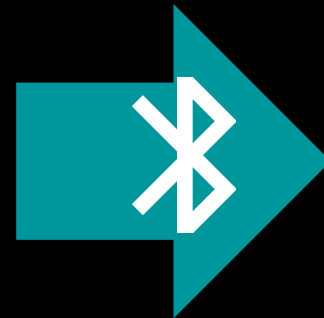


A Device
Nano 33 BLE Sensor Node

- 9-axis motion.
- accelerators
- gyroscope
- Barometers
- phase markers
- BLE payload every 50ms.



IMU, timestamp, ax, ay, az, gx, gy, gz, axx_norm, gyro_norm, phase



Computation Server

Arduino UNO Q Edge Hub

- Machine Learning
- Post/Preprocessing



InfluxDB Dashboard
for Gait Analysis

Gait Analysis

accelerators, gyroscope, Barometers



OpenRouter

google/gemini-2.5-flash

Text To Speech API



Actions/ Improvement

Rehabilitation
Coach AI Agent



Painpoint

Solution

DEMO

Architecture

Opportunity

How We Make Money

- SaaS**

Monthly Subscription
Per-device software fee. Recurring revenue, low churn.
- DaaS**

Device-as-a-Service
Hardware bundled with plan for B2B facilities. Zero upfront for buyer.
- Sale**

Outright Purchase
One-time hardware + annual software license for enterprise.

SAAS PRICING / DEVICE / MONTH

STANDARD

\$499
per device / mo

- ✓ Live cadence monitoring
- ✓ Voice coaching
- ✓ Mobile app access

PROFESSIONAL

\$799
per device / mo

- ✓ Everything in Standard
- ✓ Caregiver dashboard
- ✓ Trend reports & alerts

ENTERPRISE

\$1,299
per device / mo

- ✓ Fleet management
- ✓ API & EHR integration
- ✓ Dedicated support SLA

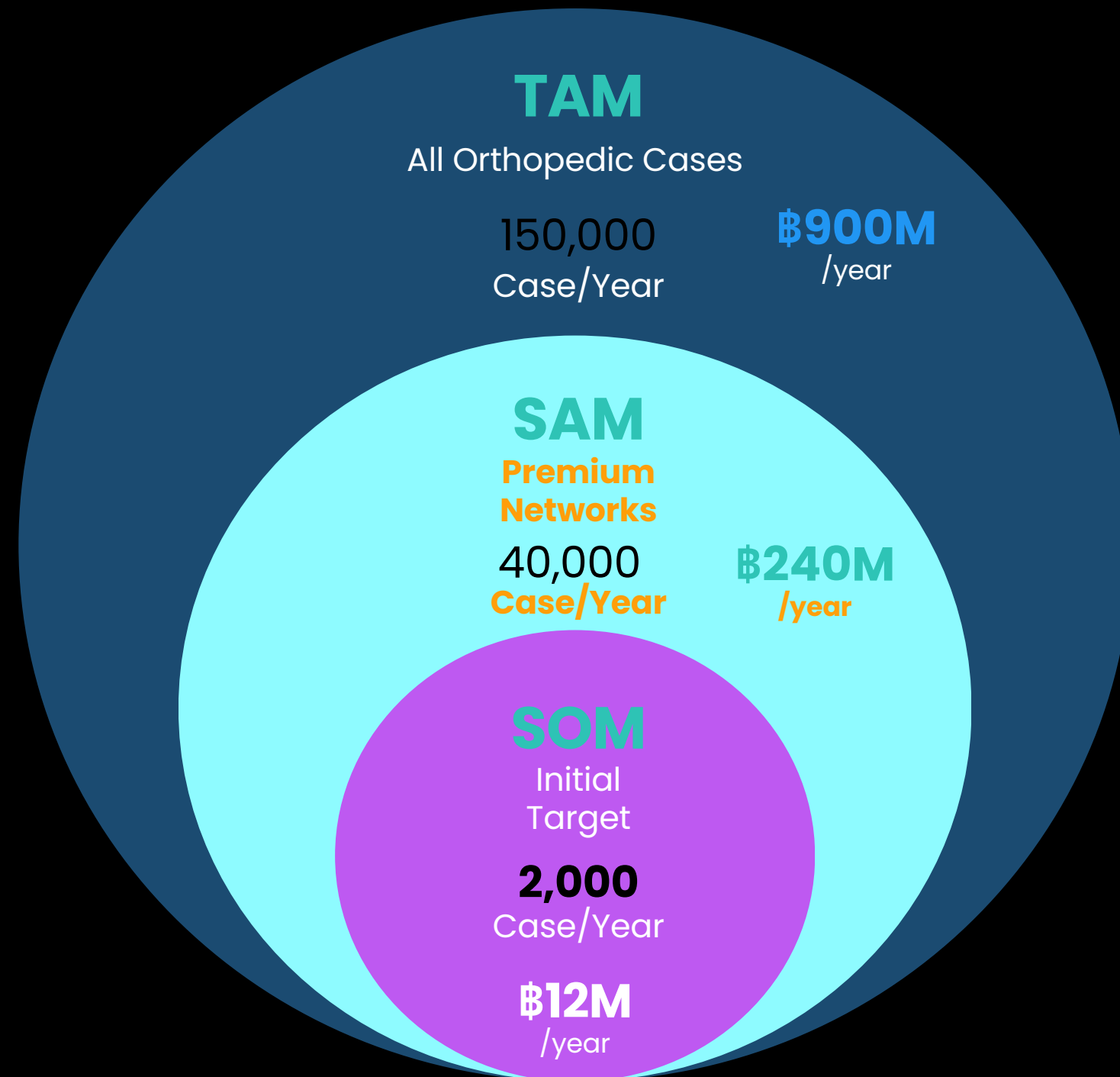
\$613,800
ARR — Standard tier, 100 devices

\$973,800
ARR — Professional tier, 100 devices

\$1.56M
ARR — Enterprise tier, 100 devices

TAM / SAM / SOM

Focused on orthopedic rehab — a validated, high-value clinical need.



SEGMENT

TAM

SCOPE & AUDIENCE

All orthopedic trauma, joint replacements & diabetic foot cases nationwide requiring weight-bearing control

CASES/ YR

150,000+

REVENUE

\$900M

SAM

Private hospital networks (BDMS) & premier university hospitals — Siriraj, Chula, Ramathibodi

40,000

\$240M

SOM

5–10 Bangkok hospital B2B2C partnerships. 5% SAM capture over 3–5 years. Early adopter phase.

2,000

\$12M

Global fall prevention market: \$38B by 2030 · 800M people aged 65+ globally by 2025

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Mission to the moon

Our 3-stage roadmap to conquer the MedTech market, starting with top-tier premium hospitals.

YEAR 1

The Beachhead (ยึดหัวหาด)



TARGET SEGMENT

3 - 5 Premium Private Hospitals
(e.g BDMS, Bumrungrad)



IMPLEMENTATION

B2B Pilot Project + 100 Clinical
validation cases

KEY MILESTONE

**5 Pilot Hospitals |
Clinical Proof**

YEAR 3

Scaling SAM (ขยายสู่ภูมิภาค)



TARGET SEGMENT

Medical schools (Siriraj, Chula) &
Premium nursing homes



IMPLEMENTATION

EHR Integration + Enterprise fleet
deployment

KEY MILESTONE

5% SAM Capture | \$12 M APP

YEAR 5

Global Vision (บุกตลาดโลก)



TARGET SEGMENT

SEA Expansion (Singapore,
Malasia) & New clinical segments



IMPLEMENTATION

Global medical distributors + US
FDA / CE Mark approval

KEY MILESTONE

**SEA Expansion | US FDA
Certified**

Competition Landscape

Our Strategic Positioning: High Clinical Rigor, Low Hardware Friction, and Proactive AI Intelligence.



MOON WALK

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CANE



WALKER
THANK YOU
Q & A



CRUTCH



PROSTHETIC LEG



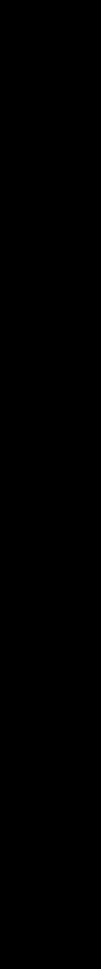
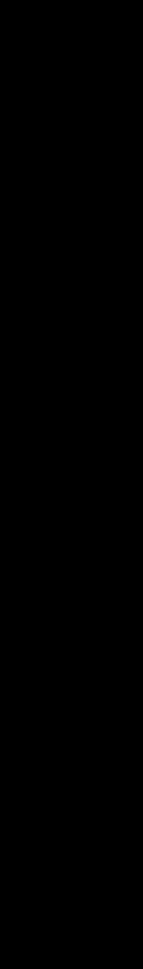
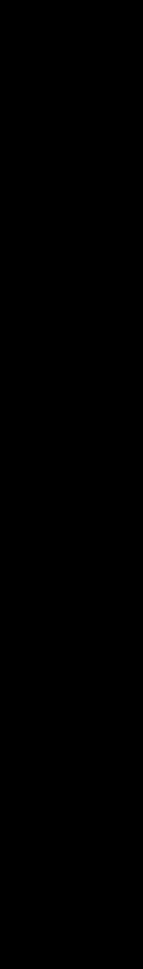
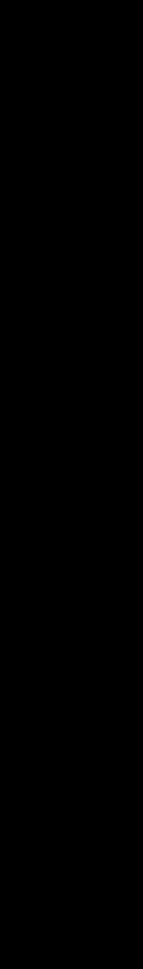
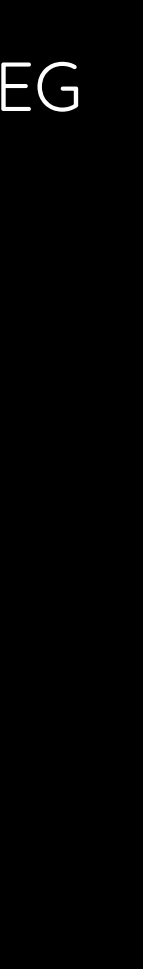
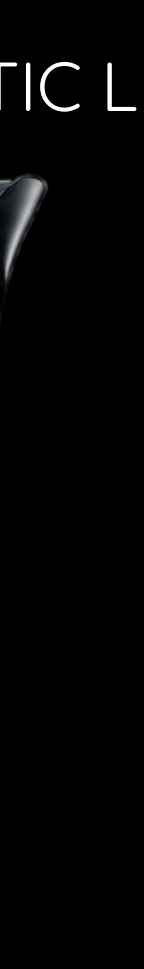
MOON WALK

Walk with your own Rhythm

CANE



WALKER



CRUTCH

PROSTHETIC LEG



MOON WALK

Walk with your own Rhythm

THANK YOU

Q & A



APPENDIX

The 3 Co-Equal Pillars of Moon **Walk** Recovery

Three distinct signals that characterize the limp, measure cane dependence, and track retraining - all calculated 100% relative to the patient's own baseline.



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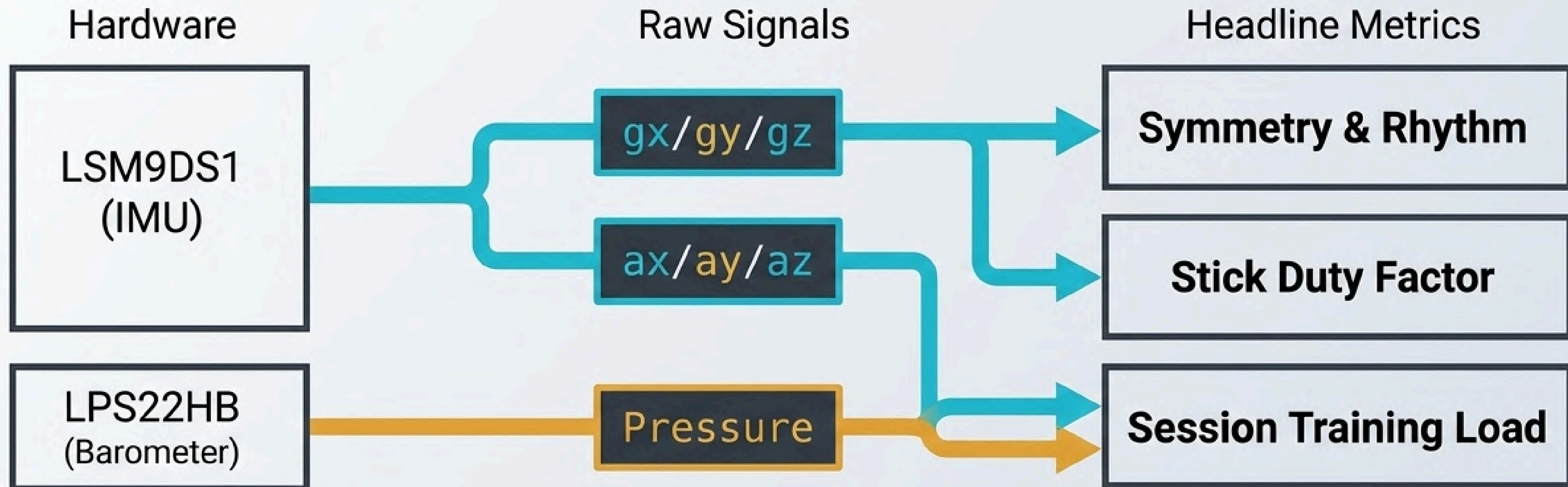
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Synthesis: The Traceability Matrix

Moon Walk extracts a 360-degree view of progressive optimal loading from an aggressively minimal hardware footprint. By eliminating fragile multi-FSR arrays and relying on a single IMU paired with a pneumatic barometer, the raw telemetry is continuously processed via onboard DSP into three robust, distinct, and co-equal clinical indicators.



Key Partners คลินิกกายภาพบำบัด นักกายภาพบำบัด / rehabilitation specialist ผู้ผลิตหรือจำหน่าย sensor และ embedded board ศูนย์ดูแลผู้สูงอายุ Medical device distributor	Key Activities - Build reliable wearable hardware - Turn walking data into actionable insight - Camera, VLM, TTS, proximity, BLE, UART integration - User testing with cane and walker users - Non-medical positioning and privacy communication	Value Propositions เปลี่ยนไม้เท้าหรือ walker เดิมให้เป็น smart assistive device แบบ self mornitoring ด้วยการ Clip-on ติดกับไม้เท้าเดิม โดยไม่ต้องซื้ออุปกรณ์ช่วยเดินใหม่ ใช้งานอัตโนมัติ, ดู trend การเดินผ่านแอป, ได้ Wellness alert แบบไม่ใช่คำวินิจฉัยเมื่อการเดินเปลี่ยนอย่างมีนัยสำคัญ วิเคราะห์ว่าการเดินวันนี้มีแนวโน้มที่ดีขึ้นจากเมื่อวานไหม Edge AI ประมวลผลบนอุปกรณ์เป็นหลัก	Customer Relationships Trust-based relationship ผ่านการออกแบบที่ให้ความสำคัญกับ privacy Wellness reminder เช่น vibration, voice prompt, dashboard alert Caregiver sharing report เพื่อช่วยติดตามผู้ใช้ที่บ้าน Clinician-readable trend report สำหรับใช้ประกอบการพูดคุยระหว่างผู้ป่วยกับผู้เชี่ยวชาญ After sale services ให้คำแนะนำ Update และดูแลตลอดการใช้งาน	Customer Segments User : ผู้สูงอายุ อยู่ระหว่างการกายภาพ หรือ มีปัญหาการเดิน secondary user : ครอบครัว บุคลากรทางการแพทย์ นักกายภาพ ผู้จ่ายเงิน : คลินิกกายภาพบำบัด ศูนย์ดูแลผู้สูงอายุ ครอบครัวผู้ใช้
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Cost Structure ต้นทุน hardware BOM: UNO Q, Nano, sensor, camera, speaker, battery, casing ต้นทุนผลิต housing / clip-on module Cloud API cost สำหรับ vision AI และ TTS R&D cost สำหรับ firmware, edge AI, dashboard และ mobile app ค่า maintenance และ technical support ค่า marketing, demo kit และ distribution Field testing และ validation cost ค่า compliance / safety review ในอนาคต			Revenue Streams รายได้จากการขายอุปกรณ์ Moon Walk hardware kit ค่าสมาชิกรสมาชิกสำหรับ AI scene description / speaking stick feature Pay-per-use สำหรับ cloud vision AI Program bundle: อุปกรณ์ + dashboard + report Maintenance / support package สำหรับองค์กรหรือศูนย์ดูแลผู้สูงอายุ	

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8 KEY PARTNERS



โรงพยาบาลและคลินิกกายภาพบำบัด
ผู้แนะนำผู้ป่วยและใช้งานรายงาน



Cloud AI Provider (Gemini API)
สำหรับ Scene Description และ TTS



ผู้ผลิตและจัดหาฮาร์ดแวร์/เซนเซอร์
Arduino, IMU, ToF, FSR, Camera, อื่นๆ



ผู้เชี่ยวชาญด้านชีวกลศาสตร์การเดิน
ให้คำปรึกษาและยืนยันความน่าเชื่อถือ



ผู้จัดจำหน่ายอุปกรณ์การแพทย์
ช่องทางจำหน่ายและบริการหลังการขาย

6 KEY ACTIVITIES



- พัฒนาและบูรณาการฮาร์ดแวร์
เชื่อมต่อ UNO Q, Nano และเซนเซอร์
- พัฒนา Edge AI สำหรับวิเคราะห์การเดิน
Baseline, Drift Detection, Gait Metrics
- พัฒนา Vision-to-Speech Pipeline
Scene Description + Proximity Alert
- ออกแบบระบบ Feedback
Vibration, Voice, LED แจ้งเตือน
- ทดสอบและเก็บข้อมูลภาคสนาม
ทดสอบกับผู้ใช้จริงและปรับปรุงต่อเนื่อง

7 KEY RESOURCES



- ฮาร์ดแวร์: UNO Q, Nano, IMU, FSR, ToF, Camera, Audio
- อัลกอริทึมและโมเดลเฉพาะทาง
Personal Baseline, Drift Detection, Sensor Fusion
- ซอฟต์แวร์: แพลตฟอร์ม Edge, Mobile App, Dashboard
- ทีมงาน: Embedded, AI/ML, Mobile, UI/UX,
Clinical Advisor, Biomechanics Expert
- ชุดข้อมูลการเดิน (Walking Dataset)
สำหรับพัฒนาและปรับปรุงโมเดล

2 VALUE PROPOSITIONS



เปลี่ยนไม้เท้าหรือ walker เดิม
ให้เป็นอุปกรณ์อัจฉริยะ
Clip-on ง่าย ไม่ต้องซื้อของใหม่



ติดตามการเดินแบบเฉพาะบุคคล
สร้าง Personal Baseline และจับการเปลี่ยนแปลง
อย่างต่อเนื่อง



แจ้งเตือนความเสี่ยงล่วงหน้า
แจ้ง "wellness cue" เมื่อ pattern การเดิน
เริ่มแย่ลง (ไม่ใช่การวินิจฉัย)



ปลอดภัย เป็นส่วนตัว
ประมวลผลบนอุปกรณ์/เนื้อถือเป็นหลัก
ไม่ส่งข้อมูลขึ้นคลาวด์โดยไม่ได้รับอนุญาต



ช่วยรับรู้สภาพแวดล้อมรอบตัว
Speaking Stick: อธิบายภาพ + เตือนสิ่งกีดขวาง
ทำงานได้แม้ออฟไลน์



ช่วยผู้ใช้และผู้ดูแลเห็นแนวโน้ม
Dashboard และรายงาน Trend
ช่วยติดตามและตัดสินใจได้ดีขึ้น

4 CUSTOMER RELATIONSHIPS



ความไว้วางใจและความเป็นส่วนตัว
สื่อสารชัดเจน ไม่เก็บข้อมูลเกินจำเป็น



การสื่อสารแบบให้กำลังใจ
ใช้ภาษาสนับสนุน ไม่สร้างความตื่นตระหนก



แชร์รายงานกับผู้ดูแล/นักกายภาพ
ทำให้ติดตามอาการและวางแผนการฟื้นฟูได้ดีขึ้น



บริการหลังการขายและการสนับสนุน
อัปเดตซอฟต์แวร์, ดูแลอุปกรณ์, ให้คำแนะนำ

3 CHANNELS



โรงพยาบาลและคลินิกกายภาพบำบัด
ช่องทางแนะนำผู้ป่วยและใช้งาน



ร้านอุปกรณ์การแพทย์และตัวแทนจำหน่าย



ช่องทางออนไลน์และ E-commerce



Companion Mobile App (BLE)



งานแสดงสินค้า / Health Tech / Rehab Expo



Caregiver Community & Online Platform

1 CUSTOMER SEGMENTS



ผู้ใช้หลัก (Primary Users)



- ผู้สูงอายุที่ใช้ไม้เท้าหรือ walker
- ผู้ที่อยู่ระหว่างฟื้นฟูสมรรถภาพการเดิน
- ผู้ที่มีปัญหาการเคลื่อนไหว / เดินกะเผลก
- ผู้ที่มีปัญหาการมองเห็น

ผู้ใช้อรอง (Secondary Users)



- Caregiver / ครอบครัว
- นักกายภาพบำบัด
- แพทย์ / บุคลากรคลินิก

ผู้จ่ายเงิน / ผู้รับผิดชอบ (Payers)



- โรงพยาบาล
- คลินิกกายภาพบำบัด
- ศูนย์ดูแลผู้สูงอายุ
- ครอบครัวผู้ใช้

9 COST STRUCTURE



ต้นทุนฮาร์ดแวร์ (BOM)
UNO Q, Nano, IMU, FSR, ToF,
Camera, Audio, Battery, Casing



ต้นทุนการผลิตและประกอบ
PCB, Assembly, Test, Packaging



ต้นทุนบริการคลาวด์
Vision AI (Gemini API), TTS



ต้นทุนพัฒนา R&D
Firmware, Edge AI, Mobile App,
Dashboard, Cloud Pipeline



ต้นทุนการทดสอบภาคสนาม
Data Collection, Validation, Usability



ค่า Compliance / Certification
มาตรฐานความปลอดภัยและคุณภาพ



ค่า Marketing & Sales
สื่อสารตลาด, Demo Kit,
งานแสดงสินค้า, ตัวแทนจำหน่าย



ค่า Support & Maintenance
บริการหลังการขาย, อัปเดตซอฟต์แวร์

5 REVENUE STREAMS



รายได้จากการขายอุปกรณ์ (Hardware Sale)
จำหน่ายชุด Moon Walk (Sensor Node + Compute Brain + Accessories)



ค่าสมัครสมาชิกฟังก์ชัน AI (Subscription)
Speaking Stick (Scene Description), รายงานเชิงลึก, Cloud AI



Pay-per-use (Cloud AI)
จ่ายตามการใช้งาน Vision AI / TTS



รายได้จากโมเดล B2B
Leasing ให้คลินิก/โรงพยาบาล
เช่าอุปกรณ์สำหรับโปรแกรม Rehab



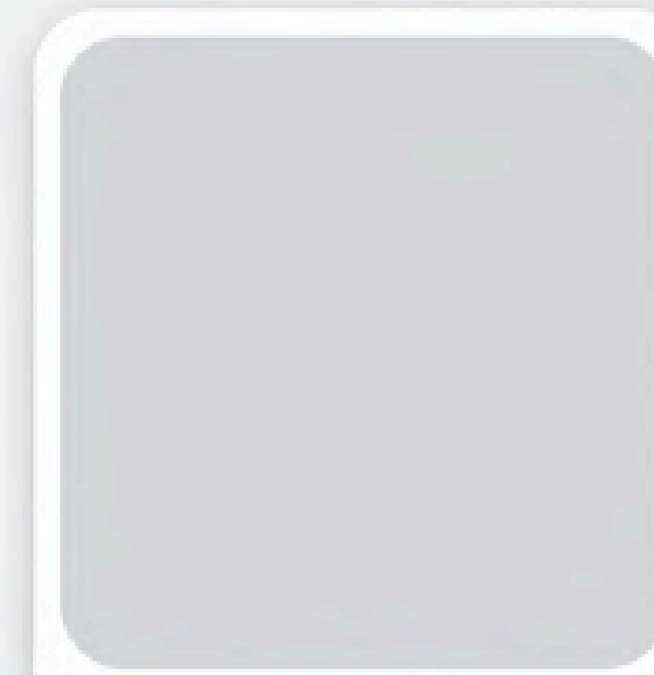
Rehab Program Bundle
อุปกรณ์ + Dashboard + Report
สำหรับโปรแกรมฟื้นฟู



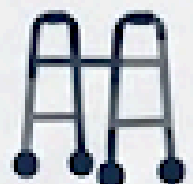
Maintenance & Support Package
บริการดูแลอุปกรณ์และซอฟต์แวร์สำหรับองค์กร

 KEY PARTNERS • Cloud AI providers (Gemini API) • Hospitals & rehab clinics • Physical therapists & medical advisors • Hardware suppliers (Arduino, sensors) • Elderly care organizations • Medical distributors	 KEY ACTIVITIES • Hardware integration (UNO Q + Nano + sensors) • Sensor fusion (IMU, FSR, ToF) • Edge AI / gait analysis logic • Dashboard & mobile app development • Feedback system (vibration, voice, LED) • Vision-to-speech pipeline • Real-world testing & safety UX design	 KEY RESOURCES • UNO Q main hub + Nano wearable node • Sensors: IMU, FSR, ToF, camera • Feedback hardware: speaker, vibration, LED • Edge AI algorithms (baseline + drift detection) • Mobile app & clinician dashboard • Walking dataset • Engineering + AI + medical advisory team	 VALUE PROPOSITIONS • Clip-on upgrade for existing cane/walker • Continuous personal gait baseline monitoring • Early mobility drift detection & wellness cues • Speaking stick: scene description + obstacle awareness • Offline proximity safety alerts • Privacy-first on-device processing • Caregiver & clinician trend insights	 CUSTOMER RELATIONSHIPS • Trust-based, privacy-first design • Non-alarming supportive feedback cues • Voice & vibration behavior coaching • Caregiver monitoring support • Clinician-readable reports • Positioned as wellness, not medical diagnosis	 CHANNELS • Hospital & clinic referral • Rehabilitation centers • Medical device distributors • Elderly care communities • Online health platforms • Companion mobile application • Health tech exhibitions & demo events	 CUSTOMER SEGMENTS • Older adults using cane or walker • Rehab patients (mobility recovery) • Users with limp or balance issues • Visually impaired users • Caregivers & family members • Clinicians / physical therapists • Hospitals, rehab clinics, elderly care centers
	 COST STRUCTURE • Hardware BOM & manufacturing • Housing & enclosure • Cloud API costs • R&D (AI, firmware, app) • Testing & clinical validation • Maintenance & support • Marketing & distribution					

MOON WALK BUSINESS MODEL V4 - COLOR PALETTE

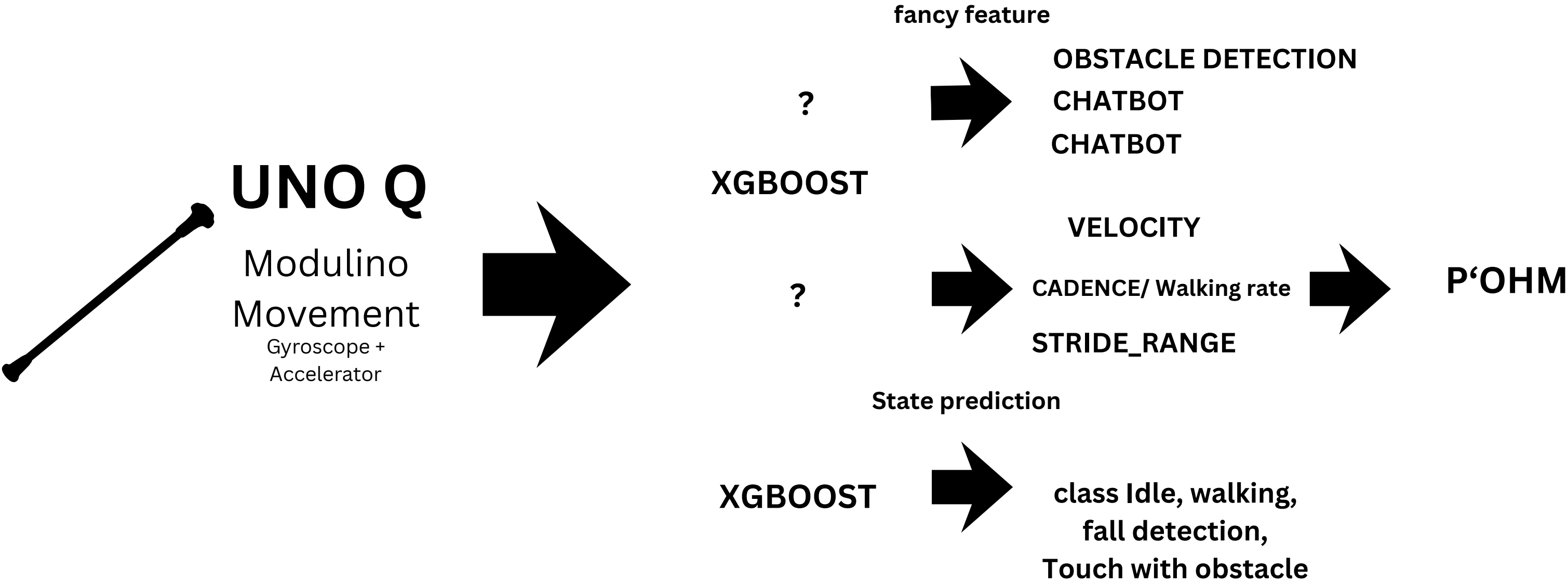
				
DEEP NAVY HEX: #1A3251	PRODUCT SLATE HEX: #3A3E41	ACCENT TEAL HEX: #32C8C2	SILVER GRAY HEX: #D4D9DD	PURE WHITE HEX: #FFFFFF
Primary text, headers, all icon outlines.	Hardware, sensor body, some device icons.	Mobility icons, specific value propositions, footer highlight.	Hardware frames, background elements, divider lines.	Module background, primary canvas color.

APPLICATION EXAMPLES

Sample icon:  

Subsidized **POC Pilot**

Pipeline



Pain Point

150,000

Orthopedic & rehab cases annually
in Thailand requiring weight-
bearing control



*"Passive walking aids
generate no data. **Moon Walk**
changes that."*

Proposed Solution

Dashboard & HCI

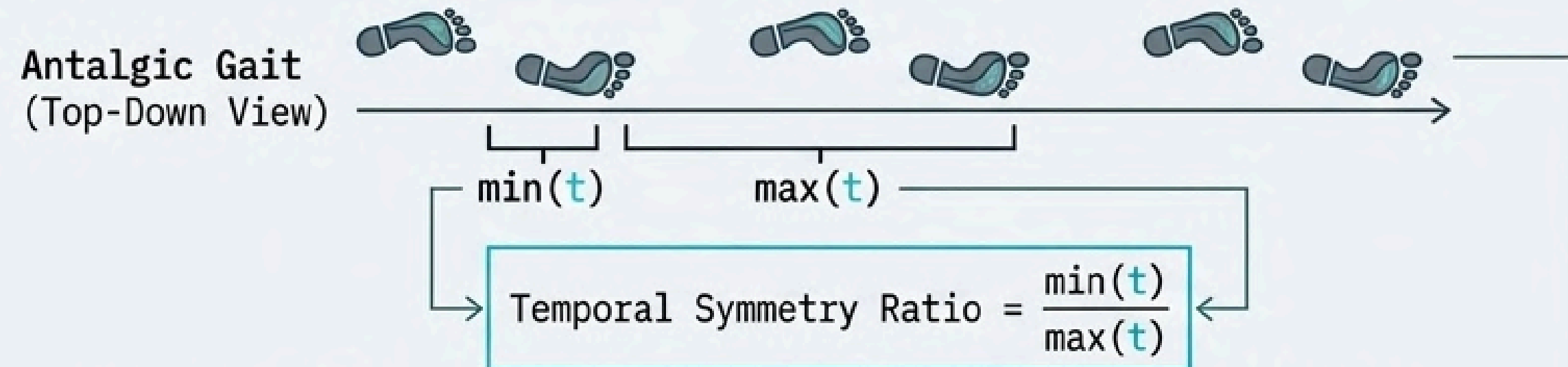
Metric 1: Symmetry & Rhythm (The Engineering)

The **headline read on a limp**: how even and how steady the walking rhythm is.
Framed as Cane-Mode Temporal Step-Time Symmetry.

Technical Definitions

- Source Signal: 6-axis IMU (gyro and accel). [Live today](#).
- Computation: The cane plants on an alternating left/right cadence.

The "Footprint Timeline" Visual



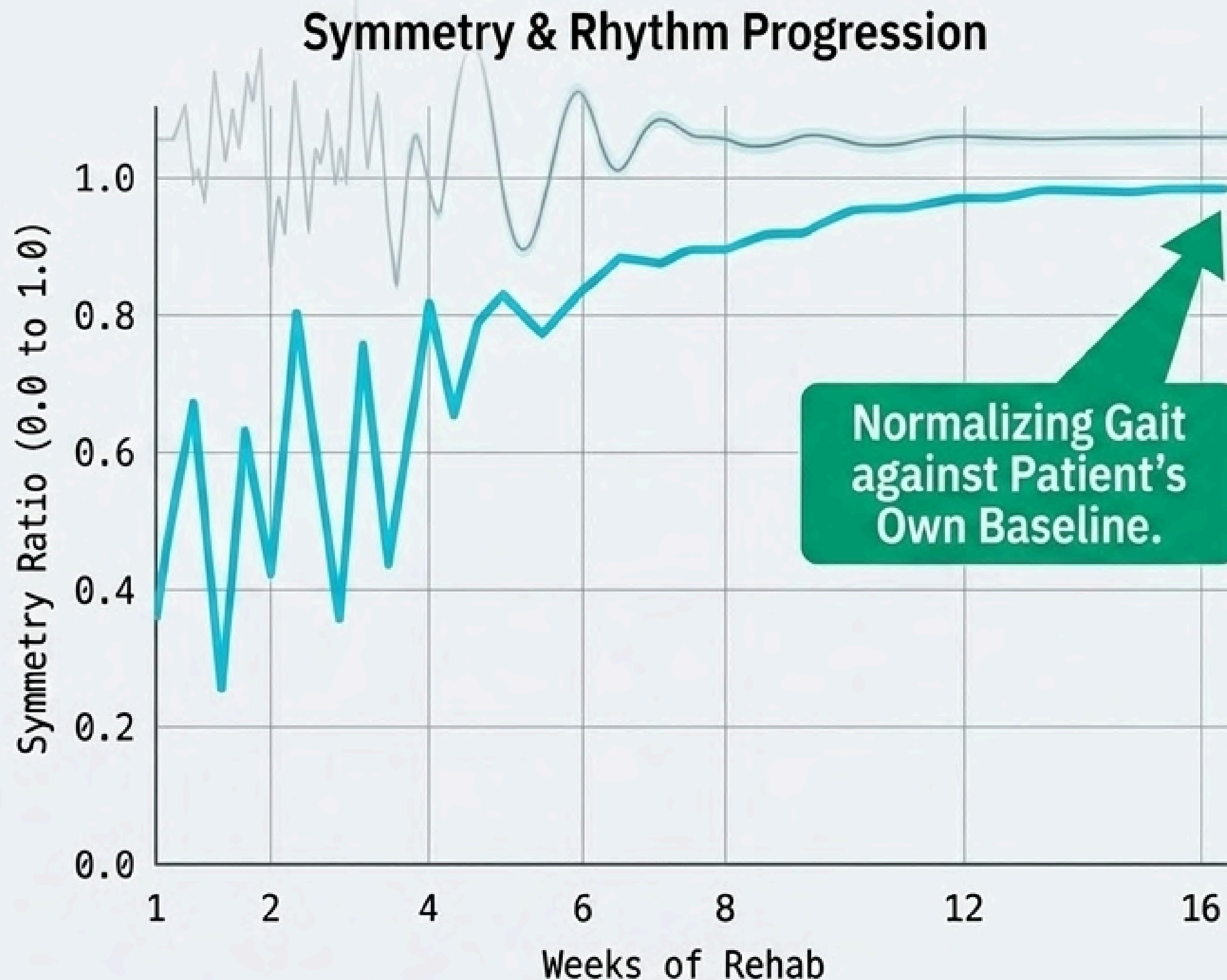
- Symmetry: The ratio of the shorter plant gap to the longer gap. (1.0 = perfectly even).

Rhythm Consistency (Step Time Variance)



- Consistency: The steadiness of the rhythm, computed as 1 minus the Coefficient of Variation (CV) of step time within each side.

Metric 1: Symmetry & Rhythm (The Clinical Evidence)



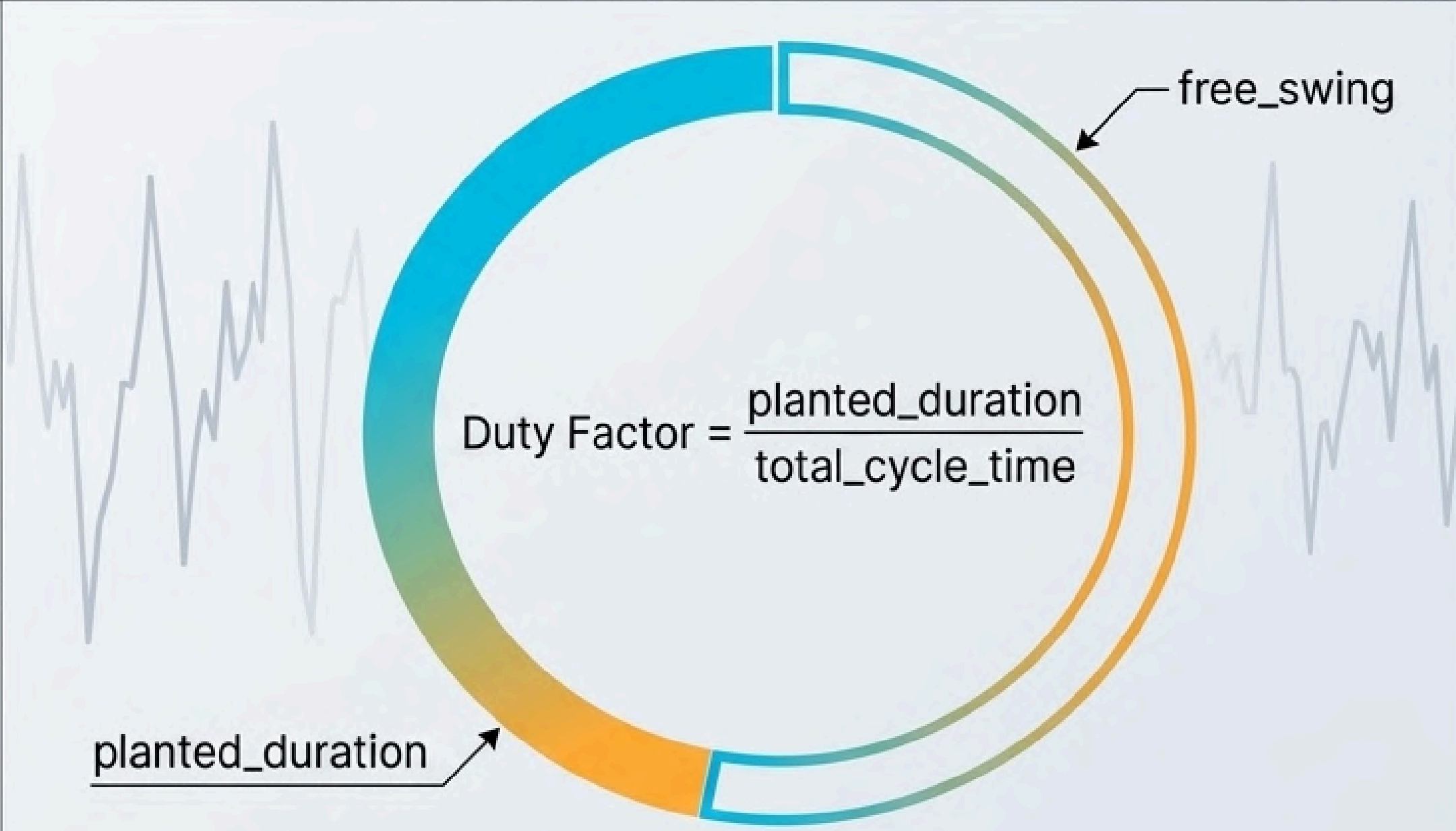
Why it matters: Asymmetry and an unsteady rhythm are the mechanical hallmarks of a limp. A trend toward symmetry and a steadier rhythm—measured exclusively against the patient's own baseline—is the clearest biological signal that the injured limb is being safely reloaded.

The Evidence Base:

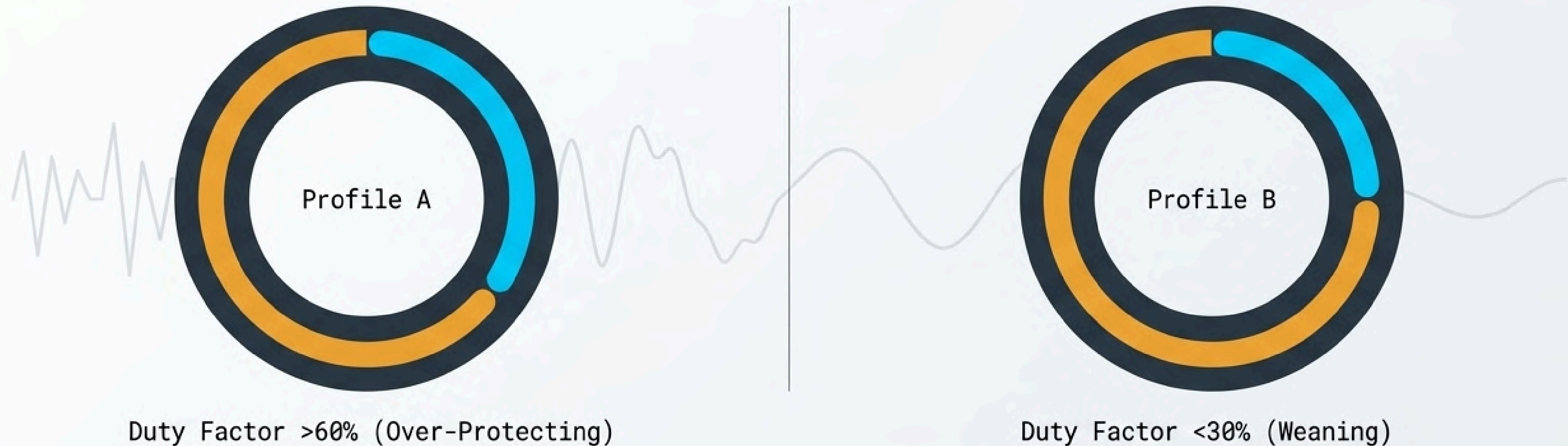
- **Musculoskeletal Validation:** Gait symmetry and walking speed are determining recovery markers after lower-limb musculoskeletal injury (Archer et al., Physical Therapy 2006). This explicitly moves the evidence base off stroke populations.
- **Standardization:** Symmetry-ratio standardization and responsiveness (Patterson 2010).
- **Rhythm Signal:** Step-time variability as the primary rhythm signal (Hausdorff 2001).

Metric 2: Stick Duty Factor (The Engineering)

A force-free proxy for cane dependence, measuring how much of each cycle the patient is actively leaning on the stick. Note: This instruments the cane, not the foot—it is not leg stance time.

Technical Definitions	The “Stick Cycle” Visual
• Source Signal: IMU primarily (stillness/plant detection) sharpened by Barometer loading edges. Live today. • Computation: Within one continuous Stick Cycle (the period from one plant to the next), Stick Duty Factor is calculated as the <code>planted_duration</code> divided by the <code>total cycle time</code>.	

Metric 2: Stick Duty Factor (The Clinical Evidence)



Why it matters:

A high or rising duty factor indicates the patient is keeping the cane planted longer to guard the injured limb. A falling trend proves they are successfully weaning off the aid.

It provides an immediate, early read on cane dependence without relying purely on force data.

The Evidence Base:

- **Cane-Sensor Viability:** Weight-bearing and cane dependence can be reliably estimated directly from sensors mounted on the cane itself (*Ballesteros et al., Sensors 2019*).
- **Cycle Segmentation:** Single-IMU gait-event detection underpinning the cycle isolation (*Salarian et al., 2004*).

Metric 3: Session Weight-Support Training Load (The Engineering)

A single, per-session figure representing how much quality loading-retraining the patient banked today.

Technical Definitions

Source Signal: **Pneumatic Handle Load (Barometer)** + **IMU step count**.

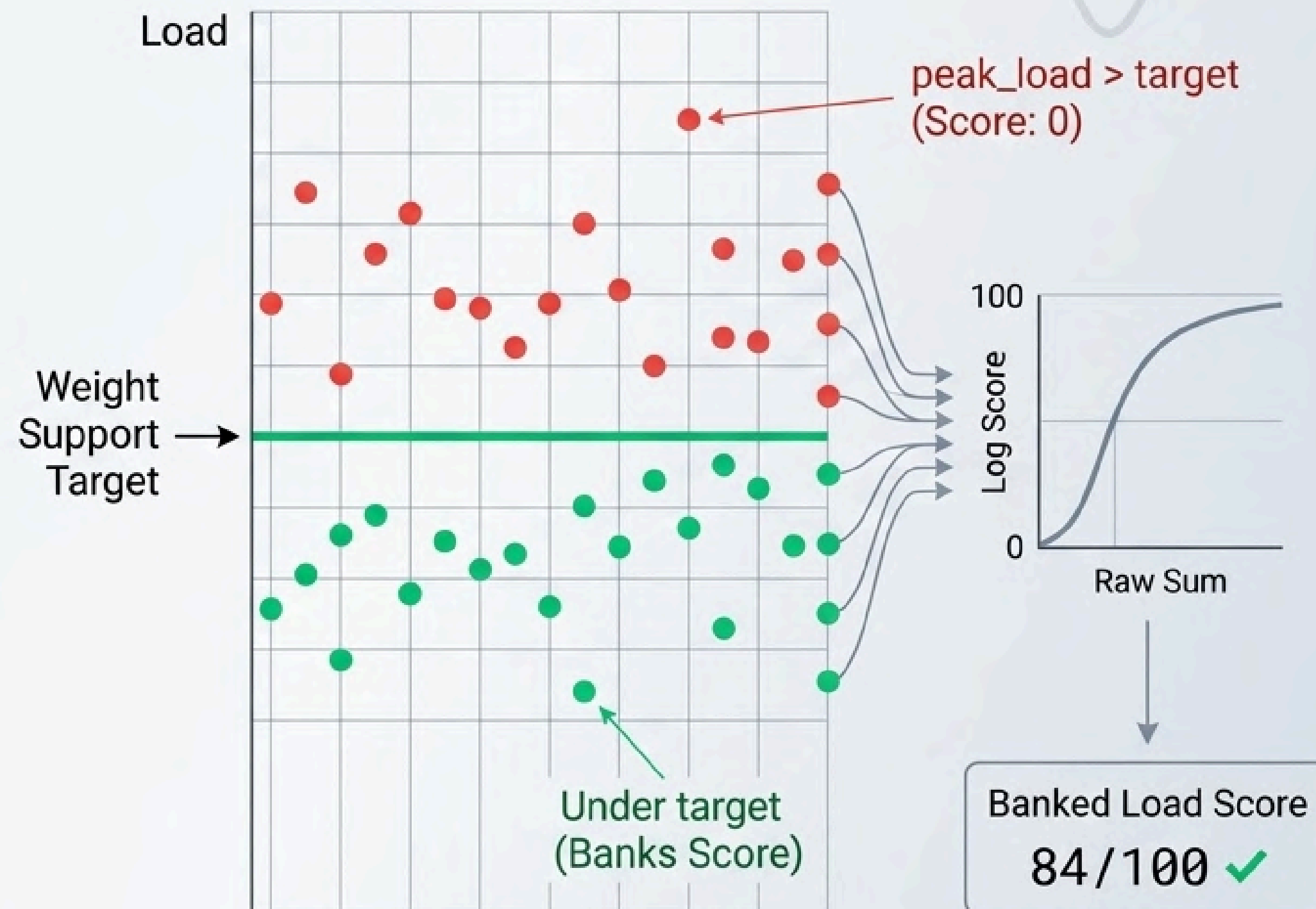
Computation: Integrating quality over the session using classic **Intensity × Volume** architecture.

- Over the limit? The step scores 0.
- Under the limit? The step earns a score based on lean reduction.

The raw sum is mapped onto a personalized, non-linear 0–100 log scale. Early gains feel highly rewarding; later gains require sustained effort.

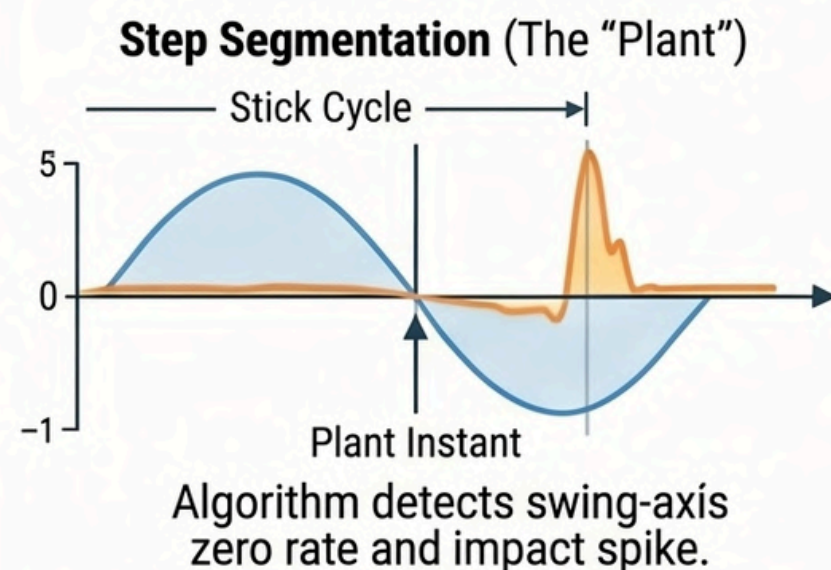
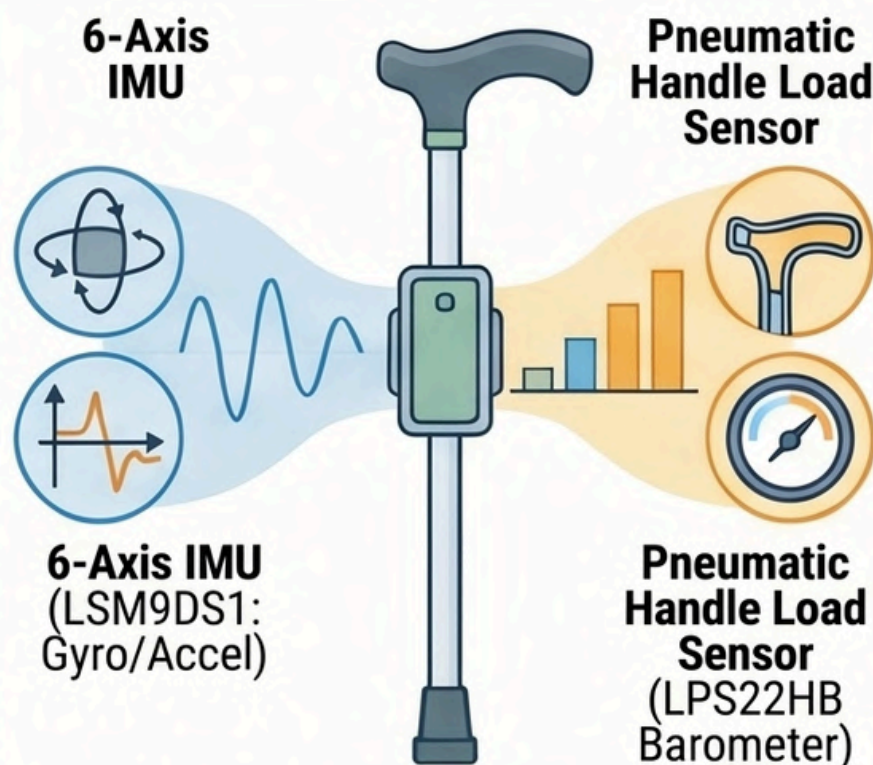
$$\text{Raw Score} = \text{Sum}(\text{Intensity} \times \text{Volume})$$

The 'Dose Grid' Visual



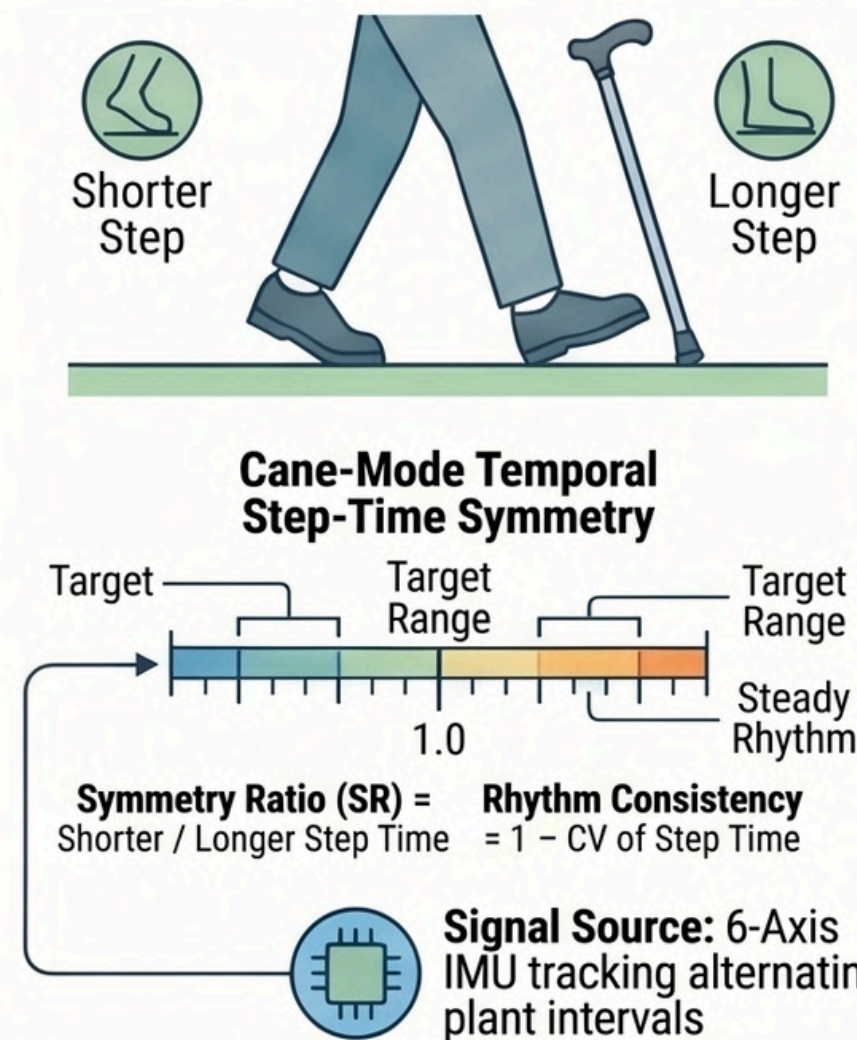
Decoding the Moon Walk: The Science of Weight-Support Biofeedback for Injury Rehab

The Sensing Foundation: Dual-Channel Sensor Suite



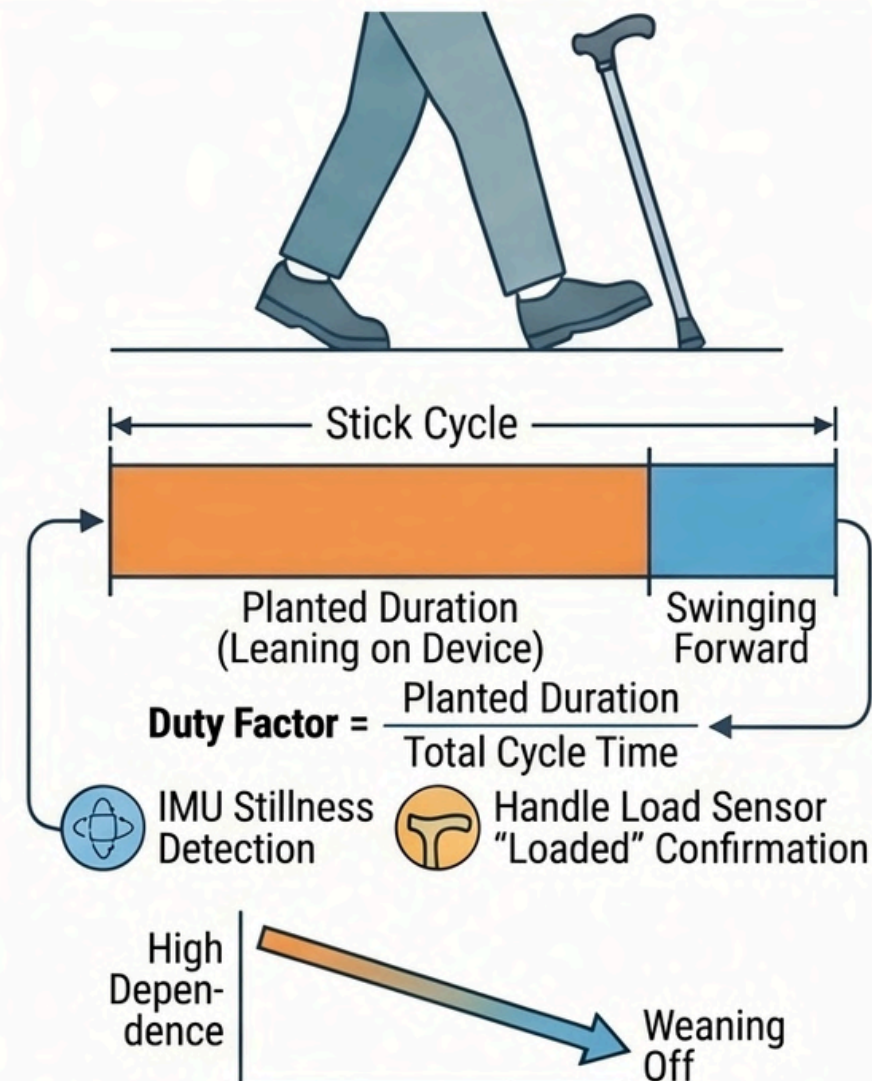
Baseline-Relative Calibration: Metrics measured against patient's own learned baseline, not population norms.

Metric 1: Symmetry & Rhythm (The Limp Signal)



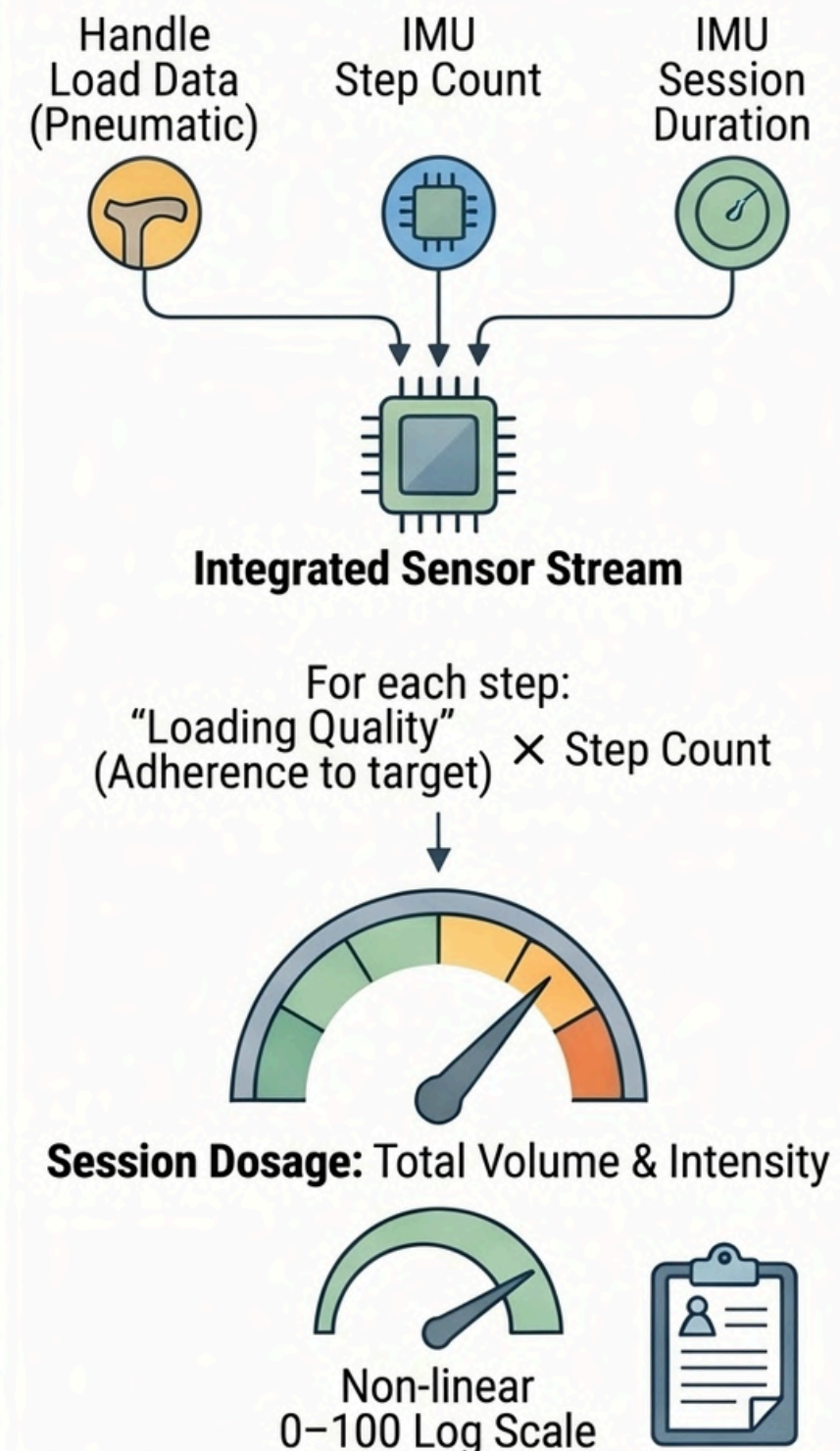
Evidence: Gait symmetry is a determining recovery marker (Archer et al., 2008); improves post-ankle sprain (Ben Moussa Zoaita et al., 2016).

Metric 2: Stick Duty Factor (Cane Dependence)



Evidence: Weight-bearing and cane dependence are reliably estimable from cane-mounted sensors (Ballesteros et al., 2019; Solarian, 2004).

Metric 3: Session Weight-Support Training Load



Evidence: Utilizes TRIMP/session-RPE structure validated in training-load science (Haddad et al., 2017) and rehab (Jung et al., 2015; Kang et al., 2020).

Clinical Evidence Reference

Metric	Primary Evidence Source	Key Finding
Symmetry	Archer KR et al. (2006)	Symmetry is a key recovery marker for musculoskeletal injury.
	Patterson (2010)	Standardized symmetry ratios are responsive to therapy.
Rhythm	Hausdorff (2001)	Step-time variability serves as the primary rhythm signal.
Duty Factor	Ballesteros et al. (2019)	Weight-bearing is estimable via cane-mounted IMUs.
Training Load	Kang et al. (2021)	WSFC protocols effectively guide loading via feedback.
	Haddad et al. (2017)	Validates the 'Intensity x Volume' construct for session load.

Multi-Sensor System

**AI / Logic
Insight**

Business Canvas

Market Opportunity

Tier 1: Temporal

Cycle Time, Cadence, Duty Factor

Tier 2: Spatial Trends

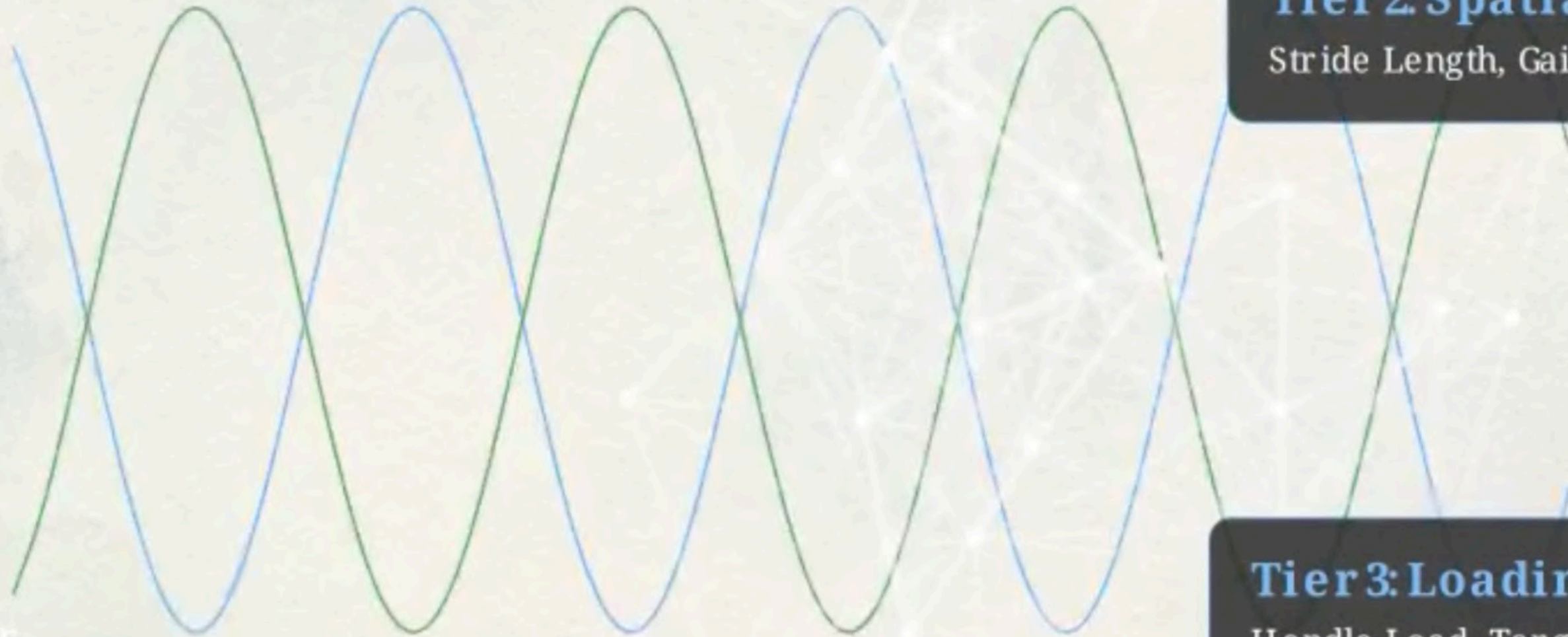
Stride Length, Gait Velocity

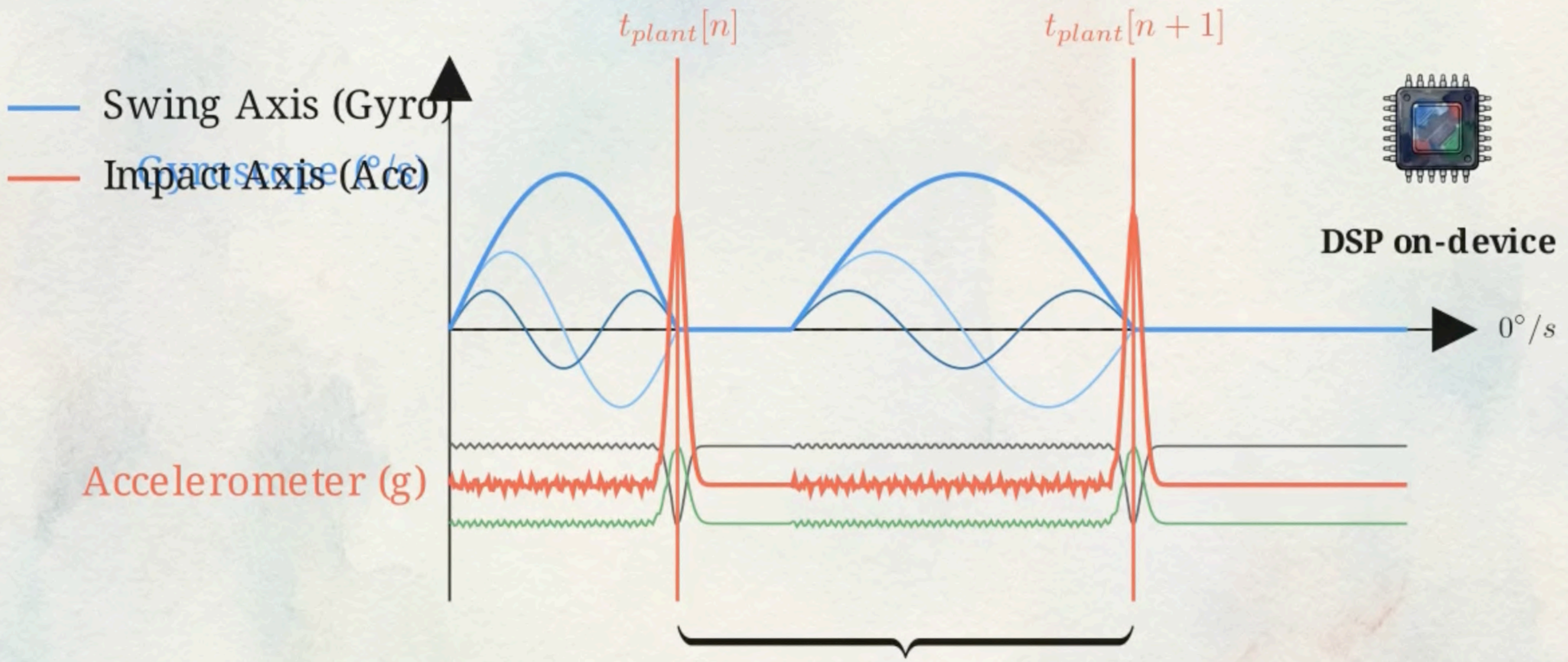
Tier 3: Loading

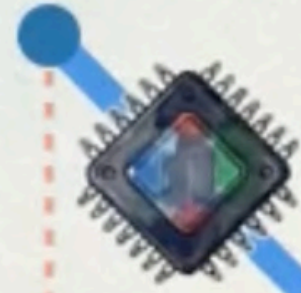
Handle Load, Target Compliance

Tier 4: Composite

Step Rhythm, Symmetry, Training Load







$$\theta = \int \omega(t) dt$$



$$\text{Stride} \approx L \cdot \sin(\theta)$$



Gait Asymmetry in Community-Ambulating Stroke Survivors

[Kara K. Patterson, MSc](#)^{a,b,c} · [Iwona Parafianowicz](#)^a · [Cynthia J. Danells, MSc](#)^a · ... · [W. Richard Staines, PhD](#)^{a,b,d} · [Sandra E. Black, MD, PhD](#)^{a,c} · [William E. McIlroy, PhD](#)^{a,b,c,d} 

[... Show more](#)

[Affiliations & Notes](#)  [Article Info](#) 



Static and cyclic performance evaluation of sensors for human interface pressure measurement

Publisher: IEEE

[Cite This](#)

 PDF



Research article [Get rights and content](#) 

Partial weight bearing after surgery for fractures of the lower extremity – is it achievable?

[Attila Vasarhelyi](#)^a  , [Tobias Baumert](#)^a, [Christoph Fritsch](#)^b, [Werner Hopfenmüller](#)^c, [Georg Gradl](#)^a, [Thomas Mittlmeier](#)^a [Show more](#) 

 Cite  Add to Mendeley  Share | [10.1016/j.gaitpost.2004.12.005](#) 



Designing a Gait Recognition Algorithm for Older Adults Using Mobility Aids: Prospective Cohort Study

[Samantha Jeane Ray](#)¹ ; [Jung In Koh](#)² ; [Amanda Mae Liberty](#)¹ ; [Tracy Anne Hammond](#)² ; [Paula Kay Shireman](#)¹ 

Real-time gait event detection for lower limb amputees using a single wearable sensor

Publisher: IEEE [Cite This](#) [PDF](#)

[H. F. Maqbool](#) ; [M. A. B. Husman](#) ; [M. I. Awad](#) ; [A. Abouhossein](#) ; [P. Mehryar](#) ; [N. Iqbal](#) [All Authors](#)

Smart Walking Aids with Sensor Technology for Gait Support and Health Monitoring: A Scoping Review

by [Stefan Resch](#) ^{1,2,*}  [Aya Zlrari](#) ²  [Thi Dlem Quynh Tran](#) ²  [Luca Marco Bauer](#) ² and [Daniel Sanchez-Morillo](#) ^{3,4} 

Can wearable real-time biofeedback gait training devices improve gait speed, balance, functional mobility and activities of daily living (ADL) in individuals post-stroke? A systematic review and meta-analysis of randomized controlled trials

Review | [Open access](#) | Published: 28 December 2025
Volume 23, article number 47 (2026) [Cite this article](#)
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[Feng-Yi Wang](#), [Yang Xu](#), [Laura Yu-Yan Luo](#), [Hao-Bin Liang](#), [Yi-Ping Jiang](#), [Zi-Qian Bai](#), [Mei-Zhen Huang](#), [Arnold Yu-Lok Wong](#), [Lin Yang](#), [Mingming Zhang](#), [Yong-Hong Yang](#) & [Christina](#)

30 May 2021: Clinical Research

Walking Training with a Weight Support Feedback Cane Improves Lower Limb Muscle Activity and Gait Ability in Patients with Chronic Stroke: A Randomized Controlled Trial

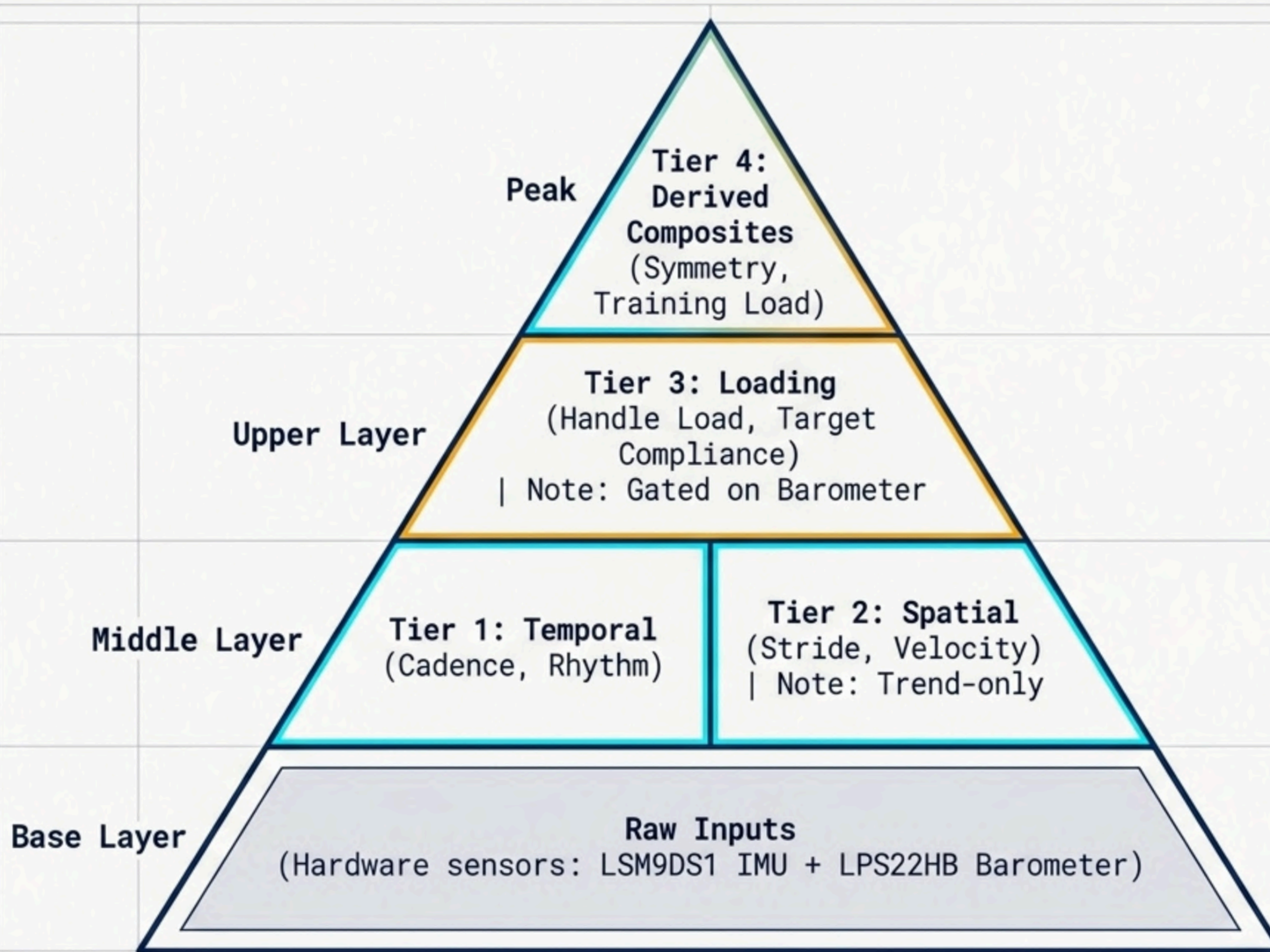
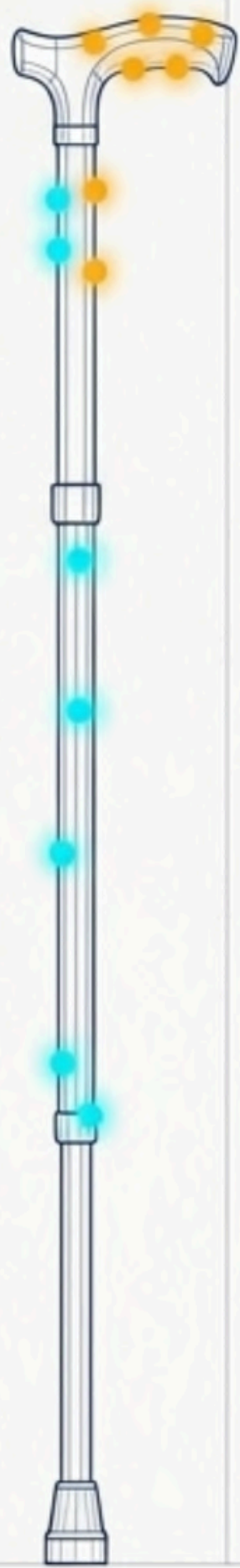
[Young Sic Kang](#) ^{1BCE}, [Gku Bin Oh](#) ^{1BCF}, [Ki Hun Cho](#) ^{1AEG*}

DOI: 10.12659/MSM.931565
Med Sci Monit 2021; 27:e931565

THANK YOU

Q & A



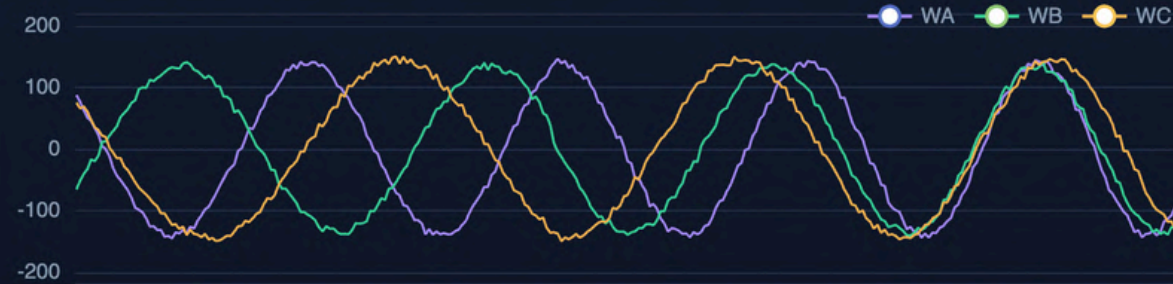


Ascending the stack increases clinical value, but requires stricter baseline personalization to maintain claim safety.

Live multi-walker view — IMU streams from each Nano 33 BLE → BLE hub → this dashboard. Walkers appear automatically as they connect.

Live signal — gyro swing (deg/s)

Raw real-time IMU stream, one line per walker (last ~6 s).



Leaderboard — current cadence

Live, all connected walkers.



Live cadence comparison (cycles/min)

Derived in the hub from Stick-Cycle detection — updates as each walker moves (last 30 s).



Training Mode — live coaching

Coaching Walker A toward their own target.



nice rhythm

target 52

now 50

cycles 29



Voice



Vibration



LED

Coaching, not correction. A wellness cue toward your own rhythm — not physiotherapy, gait correction, or medical treatment.



<https://moonwalkerscamper.netlify.app>



MOON WALK

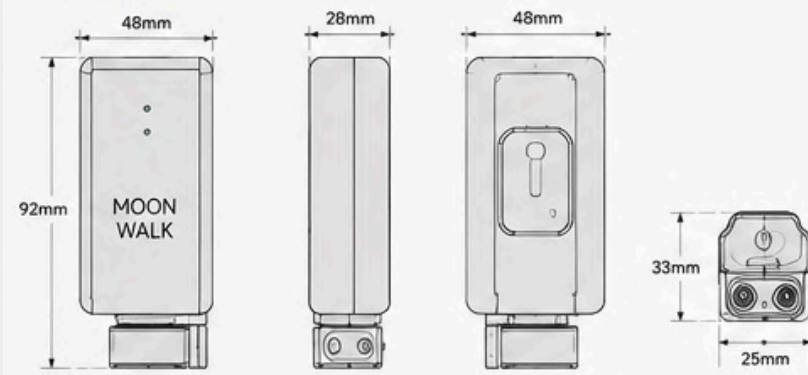
CANE SETUP – CLIP-ON SMART MODULE

Clip-on design fits most standard canes

MAIN 3D VIEW



DIMENSIONS



*All dimensions are approximate.

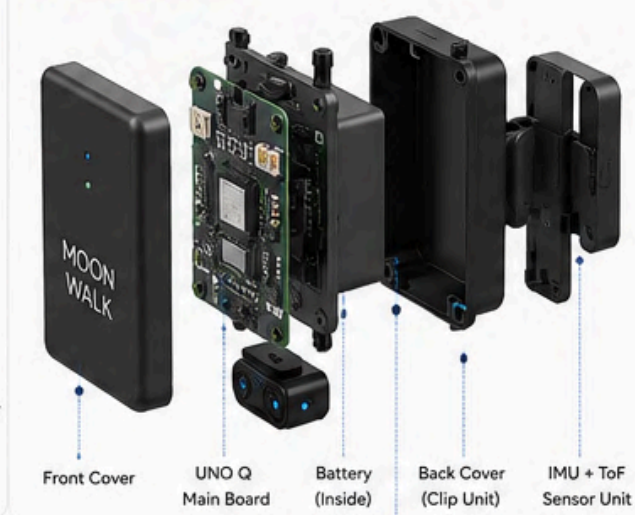
MULTI-ANGLE VIEWS



CLIP-ON MECHANISM



EXPLODED VIEW



SUITABLE CANE DIAMETER



MOUNTING EXAMPLE



IN THE BOX



Tier 1: Temporal

Cycle Time, Cadence, Duty Factor

Tier 2: Spatial Trends

Stride Length, Gait Velocity

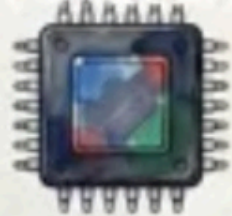
Tier 3: Loading

Handle Load, Target Compliance

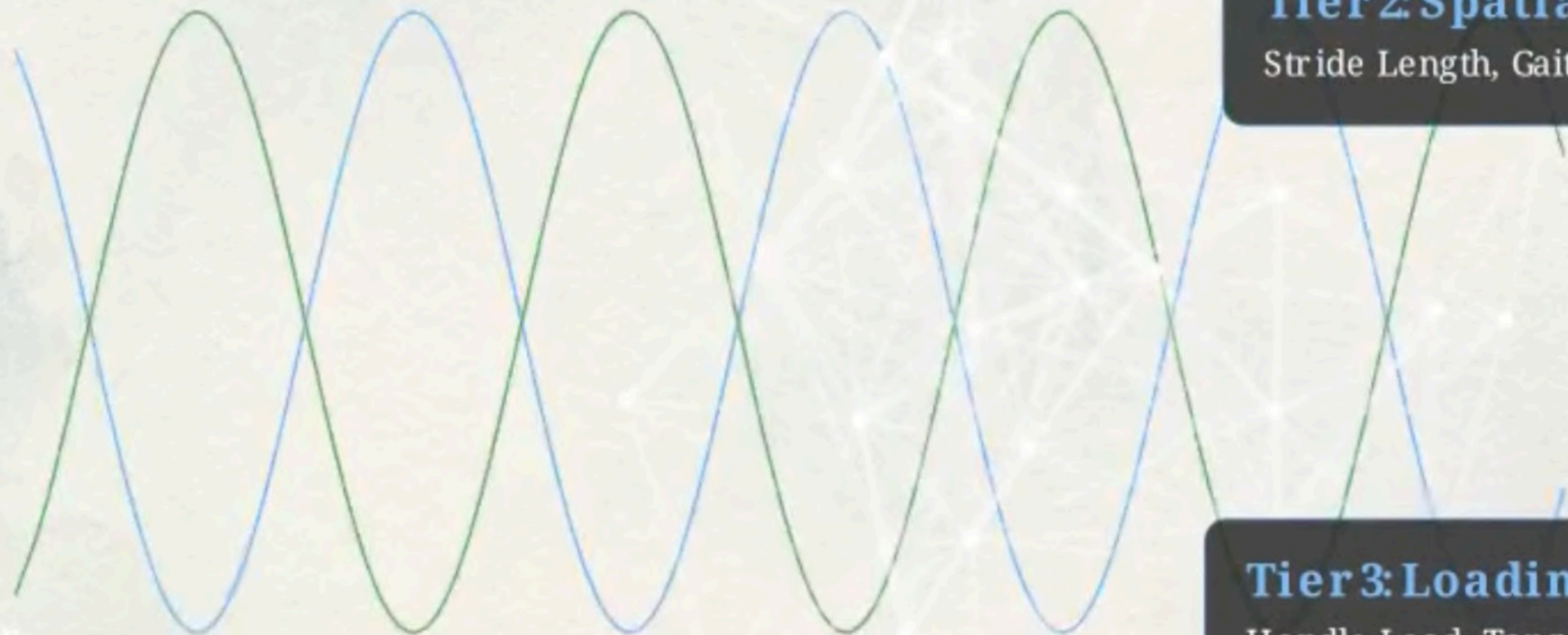
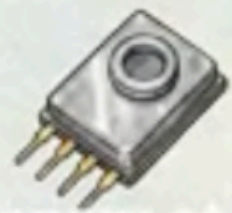
Tier 4: Composite

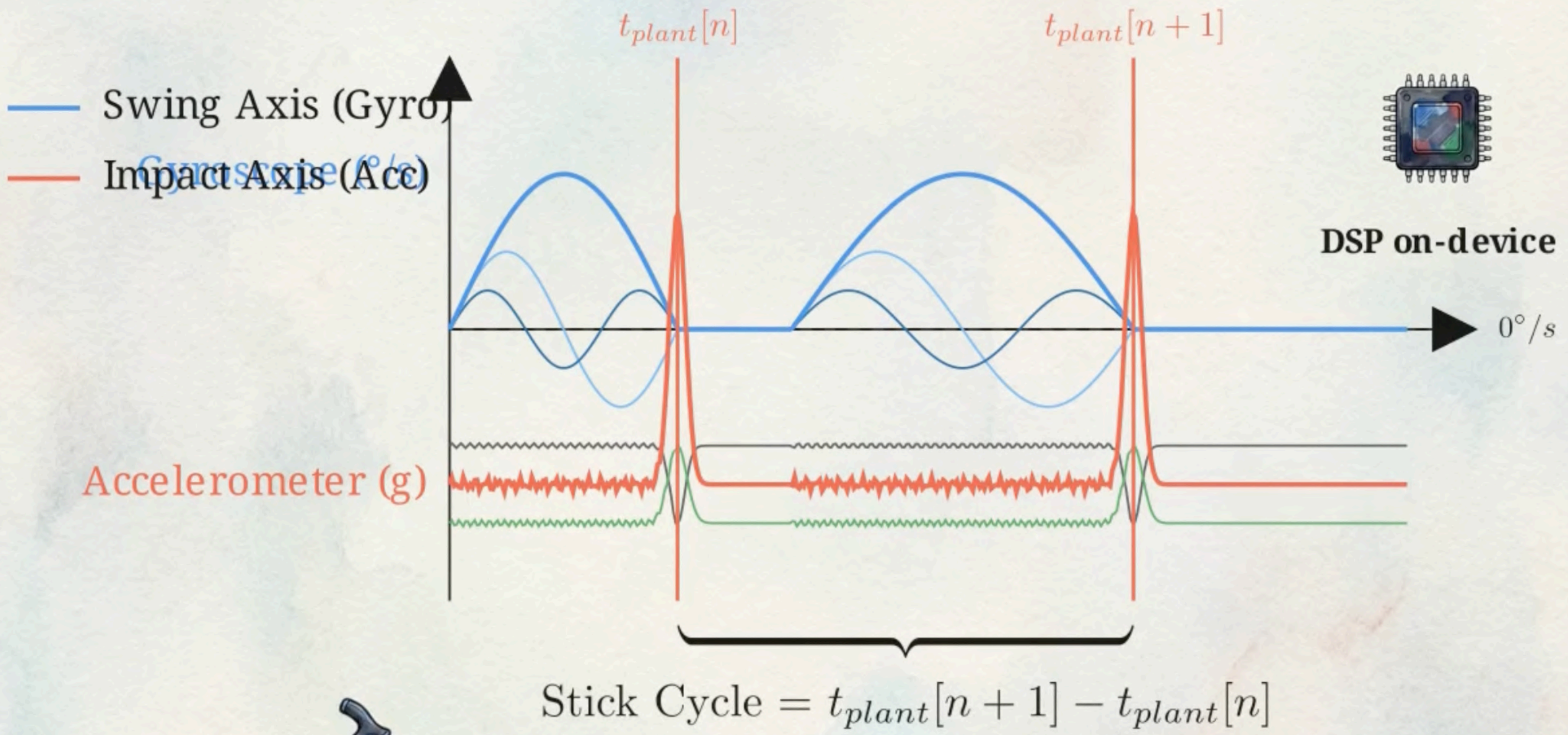
Step Rhythm, Symmetry, Training Load

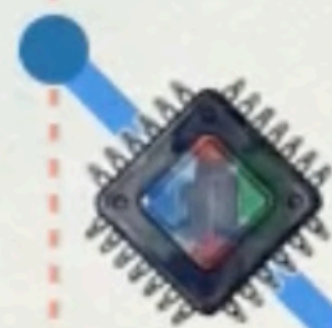
IMU
(6-axis)



Barometer
(1-axis)







$$\theta = \int \omega(t) dt$$



Stride $\approx L \cdot \sin(\theta)$



SOURCE

Mock

BLE

Scan

— scan for devices —



Connect

PHASE

STATIONARY / ZERO-VEL

acc_norm 9.76 gyro_norm 0.75

ACCELEROMETER (M/S²)

ax	-0.032
ay	0.023
az	9.755

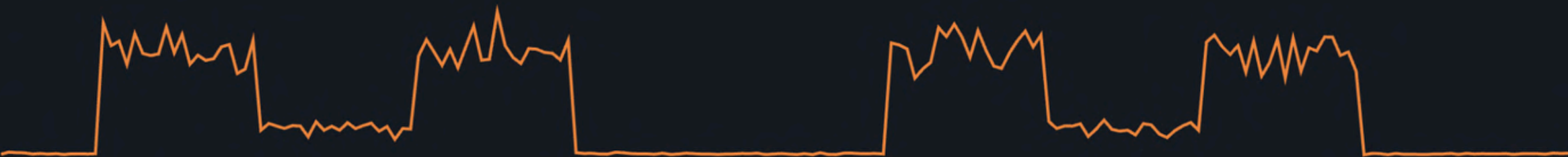
GYROSCOPE (DEG/S)

gx	0.705
gy	0.192
gz	0.189

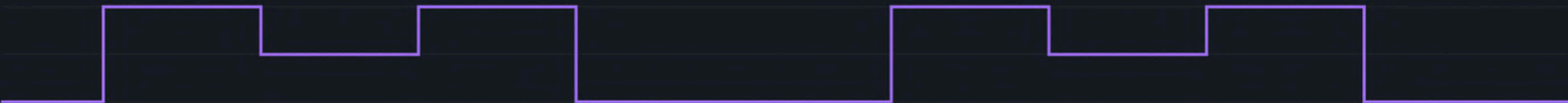
ACC_NORM



GYRO_NORM

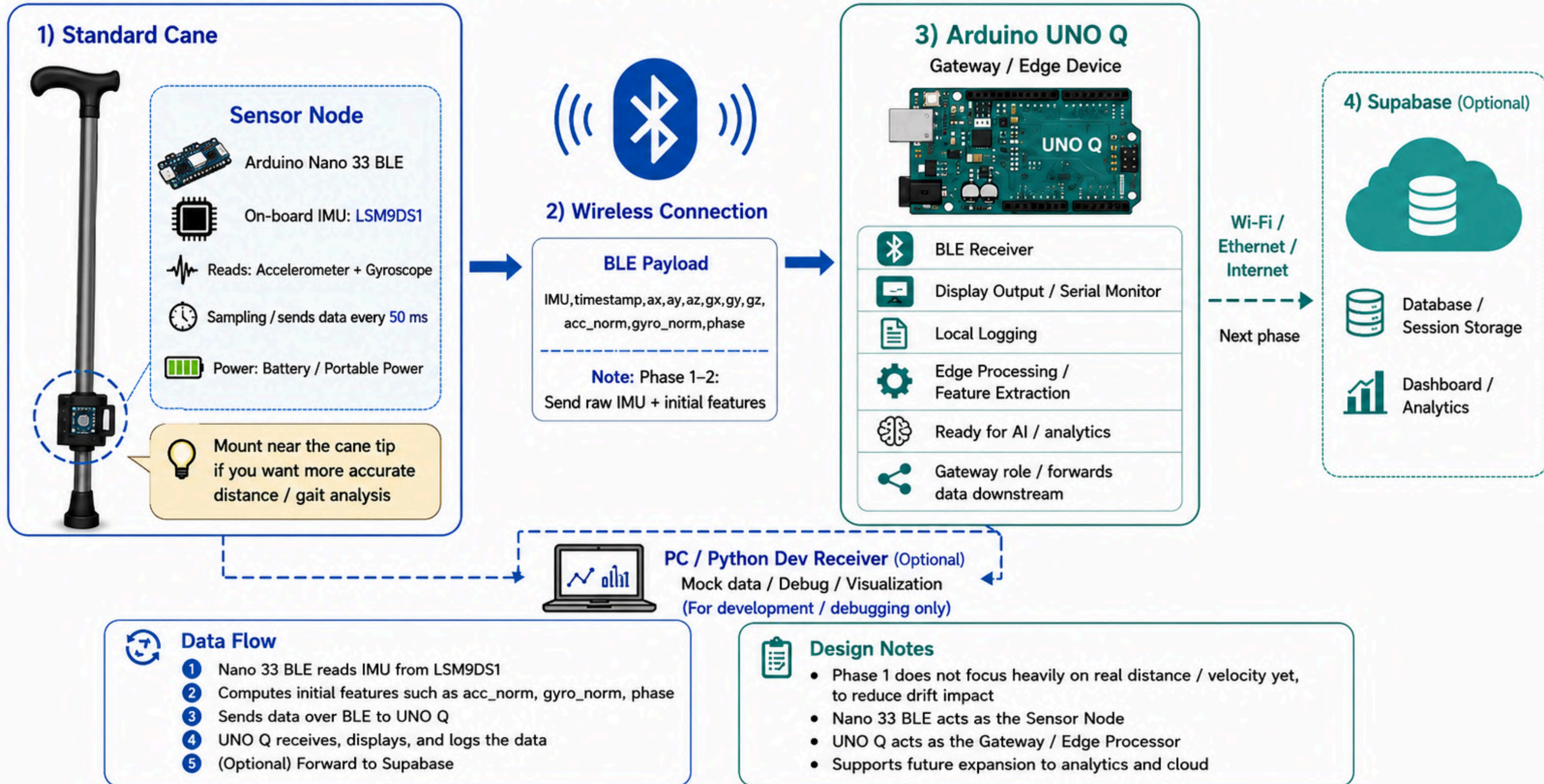


PHASE (0 UNKNOWN · 1 STILL · 2 GROUND+ROT · 3 SWING)



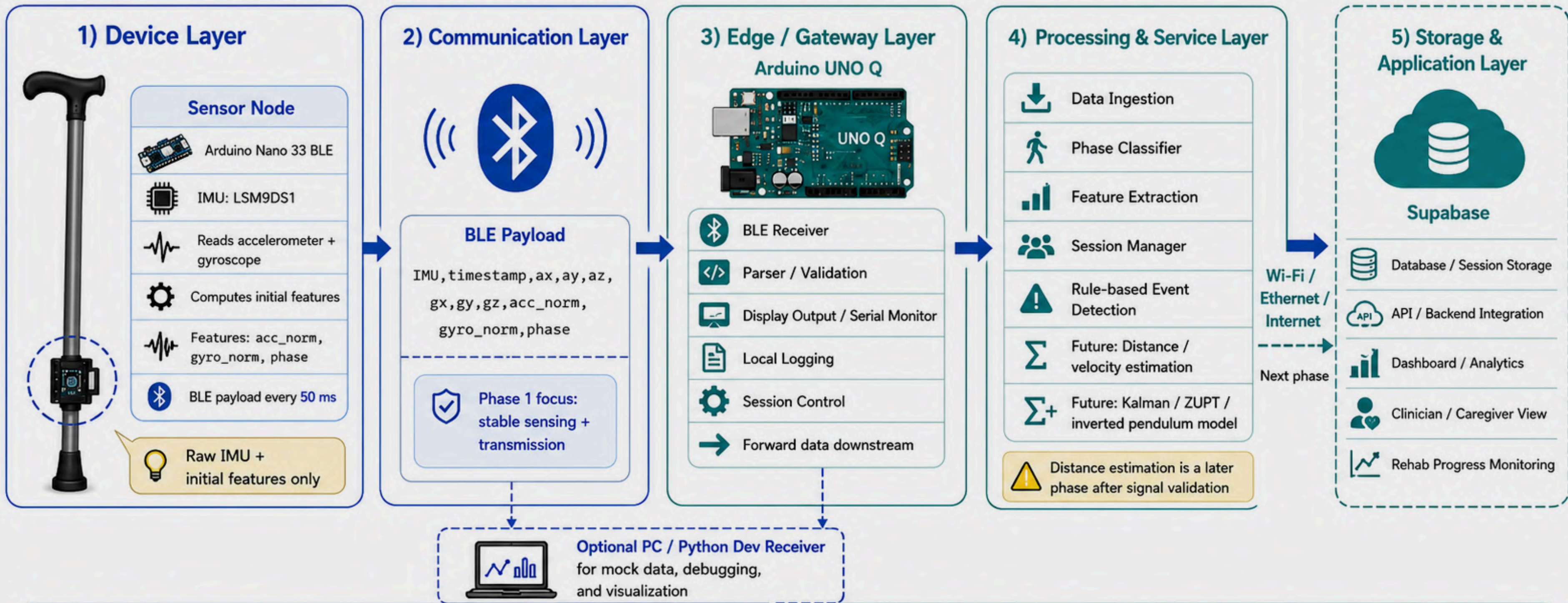
Hardware Architecture — Smart Cane IMU Prototype

Hardware-focused architecture based on our discussion: Nano 33 BLE + LSM9DS1 + BLE Gateway



System Architecture — Smart Cane IMU Prototype

End-to-end system view based on our discussion: Sensor Node → Gateway → Processing → Storage → Analytics



Data Flow

- 1 Nano 33 BLE reads IMU data from LSM9DS1
- 2 Nano computes initial features and phase labels
- 3 BLE sends payload to UNO Q
- 4 UNO Q receives, parses, displays, and logs data
- 5 Processing layer extracts higher-level session insights
- 6 Optional cloud sync to Supabase
- 7 Dashboard and analytics present outcomes

Design Notes

- Start with raw IMU + stable BLE pipeline
- Nano 33 BLE acts as the Sensor Node
- UNO Q acts as the Gateway / Edge Device
- Python can be used for mock data and debug
- Distance / velocity are not Phase 1 priorities due to drift
- Architecture supports future analytics and cloud expansion

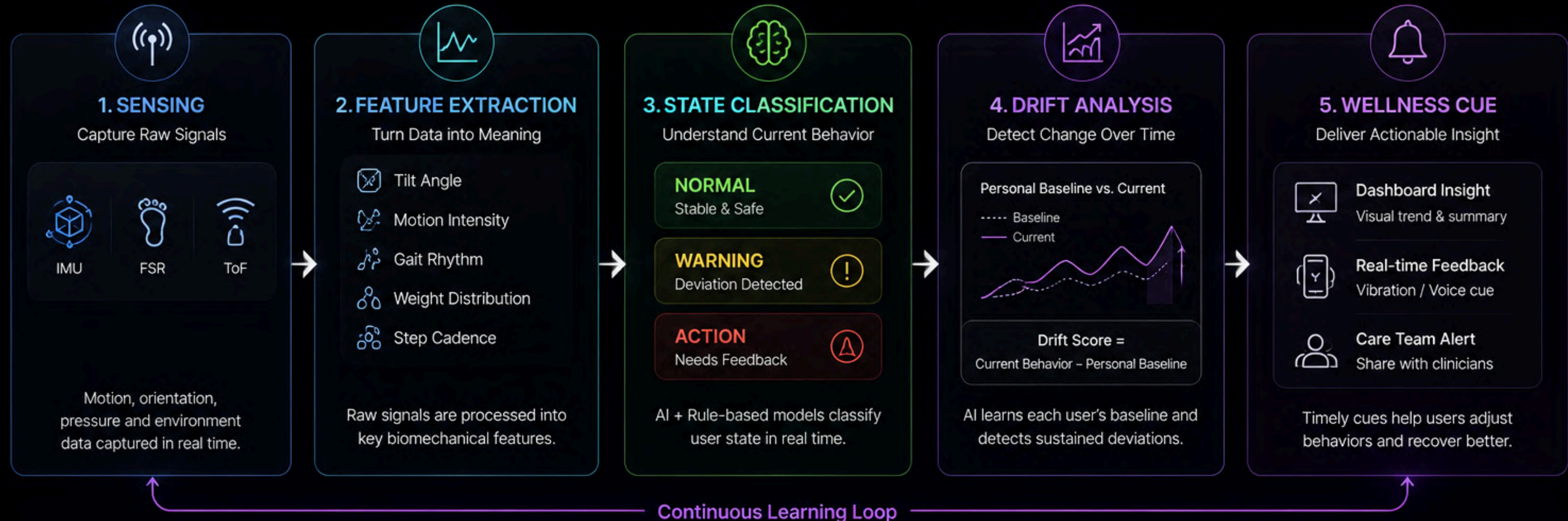
AI Logic Insight

From Motion to Meaningful Mobility Awareness

Moon Walk uses AI to learn each user's normal movement patterns, detect sustained changes, and deliver actionable wellness cues.

We don't detect problems.
We detect change.

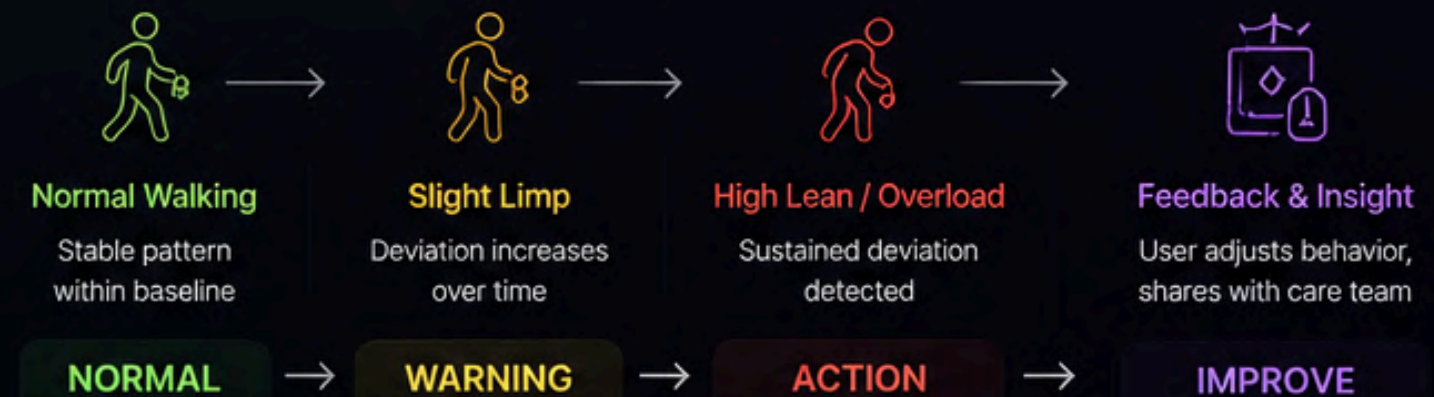
Early awareness leads to
better recovery and
better outcomes.



AI PRINCIPLES



EXAMPLE USER JOURNEY



Live signal — gyro swing (deg/s)

Raw real-time IMU stream, one line per walker (last ~6 s).



Live cadence comparison (cycles/min)

Derived in the hub from Stick-Cycle detection — updates as each walker moves (last 30 s).



<https://moonwalkerscamper.netlify.app>

Leaderboard — current cadence

Live, all connected walkers.



Training Mode — live coaching

Coaching **Walker A** toward their own target.



ease the pace

target 52

now 57

cycles 42



Voice



Vibration



LED

Coaching, not correction. A wellness cue toward your own rhythm — not physiotherapy, gait correction, or medical treatment.



Help People Move Safer

Real-time awareness & feedback



Drive Better Decisions

AI insights for caregivers & clinicians



Generate Scalable Revenue

Recurring SaaS + B2B models

Moon Walk

Clip-on smart module
for any cane or walker



Multi-Sensor

IMU • FSR • ToF



AI-Powered Insight

Detect changes from
personal baseline



Real-Time Feedback

Vibration • LED • Voice

SaaS Subscription (Per Device / Month)

MOST POPULAR

STANDARD

Essential Monitoring

499
THB

- ✓ Real-time Status (upright / leaning / moving)
- ✓ Daily Activity Summary
- ✓ 30-Day Data Retention



Best for
Small Clinics &
Nursing Homes

PROFESSIONAL

AI Insights & Reports

799
THB

- ✓ Everything in Standard
- ✓ Mobility Trend Analysis (AI)
- ✓ Weekly Automated Reports
- ✓ 1-Year Data Retention



Best for
Medium-Large Care
Facilities

ENTERPRISE

Integration & Scale

1,299
THB

- ✓ Everything in Professional
- ✓ Open API & Webhooks
- ✓ Custom Dashboard (White-label)
- ✓ Dedicated SLA Support (99.9% Uptime)
- ✓ Platform Fee Waived



Best for
Hospitals & Large
Organizations

Platform Base Fee: 15,000 THB / year

Includes Central Dashboard, Multi-device Management, Cloud Infrastructure (Waived for Enterprise)

B2B Deployment Models



1. Device-as-a-Service (DaaS)

OpEx – Monthly Subscription (24-Month Contract)



No Upfront Cost



Includes Hardware



Payback < 7 Months

Ideal for: Nursing Homes, Clinics



2. Outright Purchase + SaaS

CapEx – Own the Hardware



Hardware Ownership



Annual SaaS License



Lower TCO Long-Term

Ideal for: Hospitals, Large Organizations



Financial Impact

(100 Devices)

Tier 1 (Standard)

613,800 THB / year

~2.5x vs traditional model

Tier 2 (Professional)

973,800 THB / year

~4x vs traditional model



Recurring. Scalable. Profitable.

Turn hardware into high-margin
data-driven revenue.

TAM / SAM / SOM

Addressing a critical clinical need in Thailand's orthopedic & rehab market

THE CLINICAL NEED

Patients must follow weight-bearing restrictions.
Today, compliance is a blind spot.



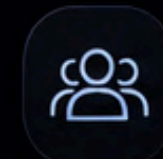
Orthopedic Trauma
Fractures from road accidents & falls



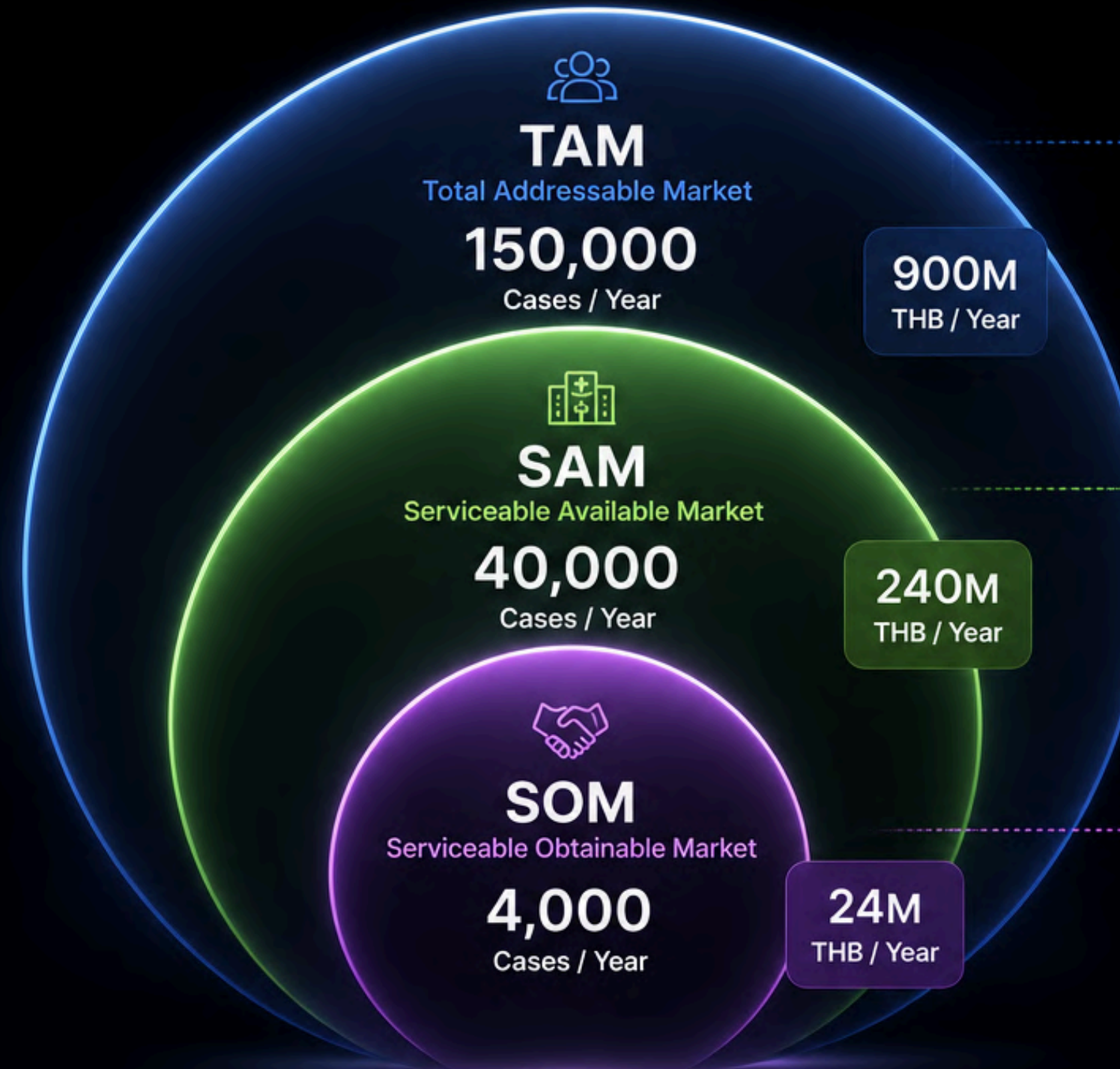
Joint Replacement
Knee & Hip replacements



Diabetic Foot
Offloading & pressure management



150,000+
Patients in Thailand every year



TAM – All Patients

All orthopedic, trauma, and diabetic foot patients who require strict weight-bearing control across Thailand.

- Road accidents & fractures
- Joint replacement surgeries
- Diabetic foot / offloading

SAM – Premium Hospitals

Patients treated in private hospitals and university hospitals with stronger ability to adopt advanced rehab solutions.

- Private hospital networks (e.g., BDMS)
- University hospitals (Siriraj, Chula, Ramathibodi)
- Higher insurance coverage & OOP spending

SOM – Initial Target

Realistic target by partnering with 5–10 leading hospital networks in Bangkok through B2B2C model.

- 10% of SAM
- Focus on early adopters
- Scalable to nationwide expansion



HIGH-VALUE MARKET
Large, growing, and underserved



RECURRING REVENUE MODEL
Hardware rental + SaaS subscription



BETTER OUTCOMES LOWER RISKS
Improved compliance
Better recovery



STRONG UNIT ECONOMICS
Reusable hardware
Multi-cycle revenue engine

We don't need the whole market.
We win the right market.

Capturing just 10% of the premium hospital segment creates **24M THB** in annual recurring revenue.



1 HARDWARE SALES (ONE-TIMES PURCHASE)

ผู้ใช้ทั่วไป

- ผู้ป่วย
- ผู้สูงอายุ
- ผู้ดูแล

องค์กร

- ศูนย์ดูแลผู้สูงอายุ
- คลินิกกายภาพบำบัด

ราคา

3000 ต่ออุปกรณ์

2 SUBSCRIPTION MODEL (RECURRING REVENUE)

Basic Plan:

Line dashboard (Basic)

State: Normal/ Warning

Basic alert

7-day Data history

Price: 0 / Month

Standard Plan:

Line dashboard (Standard)

Trend analysis 1 years

Personal baseline tracking

smart alert (Drift detection)

Caregiver Dashboard Access

Export report

Price: 150 - 300 / Month

3 LEASING MODEL (SHORT-TERM RENTAL)

Rehab bundle:

- Equipment + Dashboard + Report
- 1 - 12 months
- Small Clinic

Price: 299 - 499 / User / Month

Clinic Lease:

- Multi equipment
- Centelize dashboard
- Management systems

Start: 999 / Month

TARGET CUSTOMERS & USE CASES



ผู้สูงอายุ

ติดตามการเดินในชีวิตประจำวัน
รู้ทันความเสี่ยง



ผู้ดูแล / ครอบครัว

ดูแลคนที่คุณรักจากระยะไกล
อุ่นใจมากขึ้น



ศูนย์ดูแลผู้สูงอายุ

บริหารจัดการผู้พักอาศัย
เพิ่มคุณภาพการดูแล

